BSCS FINAL PROJECT
Software Design Specification

HASSLE-FREE: AI-POWERED CAREER ENHANCEMENT AND RECRUITMENT PLATFORM

Project Advisor

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Faculty of Information Technology & Computer Science

University of Central Punjab

University of Central Punjab

Software Design Specification

SDP Phase II

HASSLE-FREE: AI-POWERED CAREER ENHANCEMENT AND RECRUITMENT PLATFORM

Advisor: MUNEEB ALI MUZAFFAR

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Member Name	Primary Responsibility
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ALI HASSAN PERVAIZ	Cloud Deployment/Employee Dashboard

Table of Contents

Table of Contents i
Revision History ii
Previous Phases Feedback ii
Abstract iii
1. Introduction 1
1.1 Product 1
1.2 Background 1
1.3 Objective(s)/Aim(s)/Target(s) 1
1.4 Scope 2
1.5 Business Goals 2
1.6 Document Conventions 2
1.7 Miscellaneous 2
2. Overall Description 3
2.1 Product Features 3
2.2 Functional Description 4
2.3 User Classes and Characteristics 4
2.4 Design and Implementation Constraints 7
2.4.1 Academic and Institutional Constraints 7
2.4.2 Technology Stack Constraints 8
2.4.3 Hardware and Performance Constraints 8
2.4.4 Integration and Interface Constraints 9
2.4.5 Security and Regulatory Constraints 9
2.4.6 Programming Language and Design Constraints 10
2.4.7 Maintenance and Deployment Constraints 10
2.5 Assumptions and Dependencies 10
2.5.1 Assumptions 11
2.5.2 Dependencies 12
3. Technical Architecture 14
3.1 Application and Data Architecture 15
3.1.1 Class Diagram 15
3.1.2 Entity Relationship (ER) Diagram 16
3.1.3 ACTIVITY DIAGRAM: 17
3.2 Component Interactions and Collaborations 20
3.3 Design Reuse and Design Patterns 25
3.4 Technology Architecture 25
4. Screenshots/Prototype 26
4.1 Workflow 26
4.2 Screens 28
4.3 Additional Information 34
5. Other Design Details 34
6. Revised Project Plan 34
7. References 37
Appendix A: Glossary 38
Recommendation Engine**: AI module matching user profiles with suitable job listings** 38
Appendix B: IV & V Report 39

Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	CS-47
Project Title	HUSTLE FREE

Problem not clear. They are just adding each and everything in the app.

CV builder, assessment, skill learning all using AI. Use of AI has to be justified.

Similar products already exist. Scope not defined. Revise the project as there is no novelty.

Major revision required. Revise your project idea in light of comments provided. Last date to submit the revised idea form is **Thursday, September 11, 2025, before 3:00 PM** in Project Office (F-304 Building A).

FoIT&CS Projects Office
F-304
042-35880007 ext: 207
Project Office Website http://smrms.ucp.edu.pk/

F25 Proposal Submission Deadline (Major Revised Projects)

PO Project Office Sep 26
To Dr Adnan Ghafoor

Dear Students,

Prepare your Proposal Document according to the revisions made. Last date to submit the Proposal Document is **Tuesday, September 30, 2025** not later than **03:00 PM**. The proposal document should be prepared using the template available at http://smrms.ucp.edu.pk/All_Phases_Templates.html. Submit the hard copy to Project Office in F-304 (Main Building).

When submitting the hard copy of the proposal document, use the duly signed Group Allocation Yellow Page (available at http://smrms.ucp.edu.pk/Forms.html) as the first page of the submission document. Also get the proposal tape bind before submission.

FoIT&CS Projects Office
F-304
042-35880007 ext: 207

Abstract

HASSLE-FREE is an intelligent, cross-platform recruitment and career enhancement ecosystem designed to transform the interaction between job seekers and employers through AI-driven automation. Modern recruitment processes especially in developing countries often rely on manual screening, inefficient communication, and subjective candidate evaluation. HASSLE-FREE addresses these issues by integrating advanced Natural Language Processing (NLP), machine learning techniques, resume parsing, AI-based mock interviews, and smart job matching algorithms into a single unified platform.

The system enhances employability by offering comprehensive skill assessment, feedback-driven interview simulations, and automated application processes. For employers, the platform provides biasfree candidate scoring, intelligent dashboards, and data-driven decision-making tools. By bridging the gap between talent and opportunity, HASSLE-FREE aims to create a scalable, efficient, and transparent recruitment ecosystem that benefits job seekers, employers, and academic institutions alike.

1. Introduction

1.1 Product

HASSLE-FREE is designed to address the inefficiencies of modern recruitment processes by providing a unified AI-driven platform that enables intelligent resume parsing, candidate skill evaluation, and automated job matching. The system reduces manual labor for employers, enhances accuracy in candidate shortlisting, and empowers job seekers with personalized career development guidance.
The final product is a **production-ready software system** integrating AI services, mobile/web interfaces, and employer dashboards.

1.2 Background

The Pakistani job market—and many other developing markets—still relies on outdated, manual systems for candidate screening and job application management. Portals like Rozee.pk[1] offer basic job listings, but they lack intelligent features such as automated resume parsing, personalized recommendations, or interview simulations. Professional networking platforms like LinkedIn[2] provide visibility but do not offer localized insights or structured scoring systems. Meanwhile, platforms like HackerRank[3], Coursera[4], and Udemy[5] provide learning resources but operate independently of recruitment systems.

HASSLE-FREE is built to integrate all these fragmented functions into a single workflow using a cloud-based, AI-enabled infrastructure. It offers smart recommendations, automated parsing, and employer dashboards built specifically to address the recruitment challenges within Pakistan and similar markets.

1.3 Objective(s)/Aim(s)/Target(s)

The primary objectives of the system include:

- **Develop a complete AI-driven job recommendation engine** using NLP to match profiles with job descriptions.
- **Implement a transparent scoring and badge system** to help both job seekers and employers understand candidate strengths.
- **Build a real-time AI mock interview module** that evaluates user responses, confidence level, communication skills, and keywords.
- **Automate job applications across various portals**, reducing manual effort for users.
- **Provide employers with an analytical dashboard** featuring charts, candidate comparisons, filtering tools, and auto-shortlisting.

These objectives directly address the limitations of existing recruitment tools and introduce modern AI solutions.

1.4 Scope

The system encompasses a wide range of modules designed for scalability and cross-platform accessibility. The scope includes:

- **Resume Parsing & NLP-based Tokenization**
  Automated extraction of skills, experience, education, and keywords from resumes using advanced NLP algorithms.
- **Smart Job Recommendation System**
  Matches job seekers to relevant roles based on their skills, qualifications, and preferences using machine-learning ranking models.
- **Candidate Scoring Engine**
  Generates employability scores and badges based on qualifications, experience, skill test results, and personality indicators.
- **AI-Based Mock Interview System**
  Simulates real job interviews with AI-assisted questions and real-time feedback on facial expressions, tone, and content relevance.
- **Employer Recruitment Dashboard**
  Allows employers to shortlist candidates, rank profiles, and review AI-generated scoring insights.
- **Cloud-Based Deployment**
  Ensures scalability, reliability, and real-time synchronization across mobile and web platforms.

The project will be built as a **full-stack, production-level system** integrating mobile development frameworks, AI pipelines, and cloud services.

1.5 Business Goals

The commercial objectives of HASSLE-FREE include:

- Reducing hiring costs for SMEs through automated screening.
- Supporting the job market by improving job seeker readiness.
- Providing Pakistan with a modern, AI-powered recruitment ecosystem.
- Offering educational institutions tools to evaluate student employability.

Establishing a scalable model that can expand to other regions.

1.6 Document Conventions

No, this SRS document does not follow any specific typographical conventions.

1.7 Miscellaneous

No, such information that we feel, will be helpful in organizing and conduct of the project work.

2. Overall Description

2.1 Product Features

HASSLE-FREE integrates multiple AI-driven recruitment and career-development functionalities into one ecosystem. The major features include:

- **AI Resume Parsing and Tokenization**

This feature allows the system to automatically extract structured information from resumes uploaded by job seekers. Using NLP algorithms, the system identifies educational qualifications, skills, experience, certifications, job titles, achievements, and relevant keywords. It then tokenizes and categorizes the data to create a detailed digital profile. This enables accurate scoring, better job recommendations, and improved employer filtering.

- **Intelligent Job Matching Engine**

The system uses a combination of machine learning ranking algorithms and NLP-based semantic similarity techniques to match candidates with suitable job postings. Job descriptions are analyzed for required skills, experience levels, and domain relevance. Similarly, candidate profiles are evaluated using parsed resume content and manually entered preferences. The matching engine then assigns a compatibility score for each job, helping job seekers focus on highly relevant opportunities and enabling employers to receive more appropriate applications.

- **AI Mock Interview and Real-Time Feedback**

This module simulates real interview scenarios using AI-generated questions based on job roles, resume data, and industry standards. The system records candidate responses and analyzes their content quality, clarity, confidence, and relevance. Additional metrics—such as speech clarity, tone modulation, and (optional) facial expression detection—provide detailed feedback. This helps users identify weaknesses and prepare more effectively for professional interviews.

- **Candidate Scoring and Badging System**

HASSLE-FREE evaluates job seekers by calculating an employability score based on multiple factors: education history, skillset, experience, test results, interview performance, course completions, and resume strength. Candidates receive digital badges such as *"Top Skilled," "Technical Specialist," "Highly Employable,"* etc., which employers can see during shortlisting. This feature ensures transparency in evaluation and motivates users to improve their skills.

- **Employer Dashboard for Recruitment Management**

Employers have access to a dedicated dashboard that allows them to post job openings, view recommended candidates, filter applicants based on skills/experience, and analyze AI-generated scores. The dashboard visualizes data using charts and tables, helping HR teams make informed decisions. Employers can track candidate pipelines, schedule interviews, review candidate badges, and automate shortlisting based onpredefined criteria.

2.2 Functional Description

HASSLE-FREE operates as a **distributed, service-oriented AI-driven recruitment system** where functionality is delivered through coordinated interactions between frontend clients, backend services, and AI modules. Unlike Phase I (which defined *what* the system should do), this section explains *how* each major functionality is realized at design level.

At runtime, users interact with the system through a Flutter-based mobile/web interface. All requests are authenticated using Firebase Authentication and routed through Cloud Functions acting as an API gateway. Data-centric operations (profiles, jobs, applications) are handled through Firestore, while compute-intensive AI tasks (resume parsing, interview analysis, scoring) are delegated to isolated AI microservices.

The functional workflow is event-driven:

- User actions (upload resume, start interview, apply for job) trigger backend events
- Backend services invoke AI modules asynchronously
- Results are persisted and reflected back to the UI in near real-time

This design ensures scalability, fault isolation, and clear separation of concerns

2.3 User Classes and Characteristics

The HASSLE-FREE system is designed to support multiple categories of users, each having different goals, responsibilities, access rights, and interaction patterns with the system. Identifying and clearly defining these user classes is essential to ensure that system design decisions, usability considerations, and security mechanisms are aligned with user needs. Certain system requirements apply only to specific user classes; therefore, user prioritization is explicitly defined.

The user classes are divided into **favored (primary)**, **secondary**, and **supporting** user groups based on their importance to system success and frequency of interaction.

- **Job Seekers (Favored / Primary User Class)**

Job Seekers are the **most important and favored user class** of the HASSLE-FREE system. The primary objective of the platform is to improve employability, career readiness, and job placement outcomes for this group. Most system features are designed primarily around the needs of Job Seekers.

Characteristics:

- Includes students, fresh graduates, and experienced professionals
- Varying levels of technical literacy (beginner to intermediate)
- Actively seeking employment, career guidance, or skill improvement
- Rely heavily on system intelligence, recommendations, and feedback
- Frequently interact with the system

System Responsibilities for Job Seekers:

- Register and authenticate into the system
- Upload resumes in supported formats (PDF, DOCX)
- View parsed resume data and skill extraction results
- Receive AI-generated employability scores and badges
- Access personalized job recommendations
- Participate in AI-based mock interviews
- View detailed interview feedback and performance analytics
- Apply for jobs and track application status

Special Requirements:

- Highly intuitive and easy-to-use interface
- Clear explanations of AI decisions (explainable AI)
- Strong privacy and data protection for personal information
- Fast system response and real-time feedback

Priority Level: Very High (Favored User Class)

- **Employers / Recruiters (Primary User Class)**

Employers and recruiters represent the second most critical user class. Their role is to identify, evaluate, and hire suitable candidates efficiently. While not as frequent users as job seekers, employers are essential for the practical and commercial success of the system.

Characteristics:

- HR managers, recruiters, startup founders, and hiring personnel
- Moderate technical expertise
- Time-constrained users focused on efficiency
- Interested in analytics, scoring, and shortlisting tools

System Responsibilities for Employers:

- Create and manage job postings
- View AI-recommended candidates
- Filter applicants based on skills, experience, and scores

- Review employability scores and badges
- Shortlist candidates and manage hiring pipelines

Special Requirements:

- Fast filtering and ranking mechanisms
- Transparent and bias-free scoring explanations
- Secure access to candidate data
- Dashboard-oriented interface with charts and summaries

Priority Level: High

- **System Administrators (Secondary User Class)**

System Administrators are responsible for maintaining system stability, security, and compliance. They interact with the system less frequently but play a critical role in ensuring smooth operation and long-term sustainability.

Characteristics:

- Technical staff with advanced system knowledge
- Backend-focused responsibilities
- Limited interaction with user-facing features

System Responsibilities for Administrators:

- Manage user accounts and role assignments
- Monitor system performance and logs
- Manage AI model versions and configurations
- Enforce security policies and data access rules
- Handle system errors and maintenance activities

Special Requirements:

- Advanced administrative dashboards
- Role-based access control
- Detailed audit logs and monitoring tools

Priority Level: Medium

- **External Systems / Third-Party Services (Supporting User Class)**

External systems are non-human actors that interact with HASSLE-FREE to enhance its functionality. Although they do not directly use the interface, they are important for extended capabilities.

Examples:

- Job portals (e.g., LinkedIn, Rozee.pk APIs)
- Cloud-based AI and NLP services
- Notification and email services

Characteristics:

- Communicate through secure APIs
- No user interface requirements
- Depend on system availability and API stability

Priority Level: Low (Supporting User Class)

- **User Class Prioritization Summary**

User Class	Importance	Frequency of Interaction
Job Seekers	Very High (Favored)	Very Frequent
Employers / Recruiters	High	Frequent
System Administrators	Medium	Occasional
External System	Low	Automated

2.4 Design and Implementation Constraints

The design and implementation of the HASSLE-FREE system are subject to a number of constraints that limit the available technical, architectural, and operational options for developers. These constraints arise from academic requirements, technology selections, security considerations, regulatory policies, and system performance expectations. All system design decisions must comply with the following constraints.

2.4.1 Academic and Institutional Constraints

- The project must comply with **University of Central Punjab (UCP) final year project guidelines** and follow the officially provided **SDP Phase II (SDS) template**.

- The system architecture, tools, and technologies must be suitable for undergraduate-level implementation within the allocated project timeline.
- The delivered system must be maintainable and understandable by future student teams or evaluators.
- Documentation, code structure, and diagrams must adhere to IEEE-style software documentation standards.

2.4.2 Technology Stack Constraints

- **Frontend Development Constraint**
  - The client application must be developed using **Flutter** to ensure cross-platform compatibility (Android, iOS, and Web).
  - Native platform-specific development (e.g., pure Android/iOS) is discouraged to maintain a unified codebase.
- **Backend and Database Constraint**
  - Backend services must rely on **Firebase** (Authentication, Firestore, Cloud Functions, Cloud Storage).
  - Use of alternative databases (e.g., MySQL, MongoDB) is restricted unless explicitly justified.
- **AI and NLP Constraint**
  - AI modules must be implemented using **Python-based frameworks** such as spaCy, NLTK, TensorFlow, or PyTorch.
  - Models must be lightweight and optimized for **serverless or container-based deployment**.
  - Real-time AI inference must operate within cloud execution time limits.

2.4.3 Hardware and Performance Constraints

- **Client-Side Hardware Limitations**

  - The system must operate on mid-range smartphones and standard web browsers.
  - Features such as AI mock interviews are constrained by device camera, microphone quality, and processing capability.

- **Timing and Response Constraints**
  - Resume parsing and job recommendation must complete within **5 seconds** under normal network conditions.
  - AI mock interview feedback must be generated in near real-time without noticeable lag.
  - Database read/write operations must not exceed acceptable latency limits.
- **Memory Constraints**
  - Large AI models must not be executed directly on client devices.
  - Memory-intensive processing must be offloaded to cloud-based services.

2.4.4 Integration and Interface Constraints

- **Third-Party System Integration**
  - Integration with external job portals (e.g., LinkedIn, Rozee.pk) is limited to publicly available APIs or permitted access methods.
  - Web scraping or unauthorized data access is strictly avoided.
- **API and Communication Protocols**
  - All inter-module communication must use **RESTful APIs** over **HTTPS**.
  - Data exchange formats are restricted to **JSON**.
- **Parallel Operations**
  - The system must handle concurrent operations such as multiple resume uploads, interviews, and job recommendations without data inconsistency.
  - Backend services must be designed to support parallel execution in a serverless environment.

2.4.5 Security and Regulatory Constraints

- **Data Privacy**
  - The system must comply with data protection principles similar to **General Data Protection Regulation (GDPR)**, including user consent and data minimization.
  - Sensitive data such as resumes, interview recordings, and personal information must be encrypted both at rest and in transit.

- **Authentication and Authorization**
  - All users must be authenticated using Firebase Authentication.
  - Role-based access control (RBAC) must be enforced to restrict access based on user roles (Job Seeker, Employer, Administrator).
- **Secure Storage**
  - Interview audio/video recordings must not be permanently stored without explicit user consent.
  - Access to cloud storage must be strictly controlled through security rules.

2.4.6 Programming Language and Design Constraints

- **Programming Languages**
  - Dart is mandatory for frontend development (Flutter).
  - Python is mandatory for AI and NLP modules.
  - JavaScript/TypeScript may be used for Firebase Cloud Functions.
- **Design Standards**
  - The system must follow modular and layered architectural principles.
  - Standard design patterns (MVC, Repository, Microservices) must be used where applicable.
  - Code must be well-documented and readable to support maintenance.

2.4.7 Maintenance and Deployment Constraints

- The system must be deployable on **cloud infrastructure** (Firebase + AWS).
- Continuous updates to AI models must not disrupt system availability.
- Logging, monitoring, and error-handling mechanisms must be in place to support debugging and maintenance.
- Deployment pipelines must support future scalability.

2.5 Assumptions and Dependencies

This section identifies the **assumptions** made during the design of the HASSLE-FREE system and the **dependencies** on external systems, technologies, and environments. If any of them prove incorrect or change during development or deployment, the system requirements, design, or implementation may be affected. Dependencies highlight such factors on which the system relies.

2.5.1 Assumptions

The following assumptions have been made during the design and development of the HASSLE-FREE platform:

2.5.1.1 User-Related Assumptions

- Users (both job seekers and employers) will have **basic digital literacy** to operate mobile or web-based applications.
- Job seekers will upload resumes in **supported and readable formats** such as PDF or DOCX.
- Users will provide **accurate and truthful information** in resumes, profiles, and job postings.
- Users will grant necessary permissions (camera, microphone, storage) for AI mock interview functionality.

2.5.1.2 Infrastructure and Environment Assumptions

- Users will have access to a **stable internet connection** during critical operations such as resume upload, AI mock interviews, and job application submission.
- Cloud services (Firebase, AWS/GCP) will remain **available and reliable** during development and production.
- Target devices (mobile phones, laptops) will meet **minimum hardware requirements** for smooth application performance.

2.5.1.3 Data and AI Assumptions

- Resumes and job descriptions will contain sufficient structured or semi-structured data for effective NLP processing.
- AI models used for resume parsing, scoring, and interviews will achieve **acceptable accuracy** within academic and computational limits.
- Bias mitigation techniques applied to AI scoring will be sufficient to reduce unfair evaluation but may not eliminate bias completely.

2.5.1.4 Organizational and Academic Assumptions

- Project requirements and evaluation criteria defined by the university will remain **stable** throughout the project lifecycle.
- Development resources, team members, and timelines will remain available as planned.
- No major changes will be introduced late in the project that significantly alter system scope.

2.5.2 Dependencies

The HASSLE-FREE system depends on several **external components, services, and technologies**. Any change, failure, or limitation in these dependencies may impact system functionality or performance.

1. **Software and Platform Dependencies**

- **Firebase Services**
  - Firebase Authentication for user login and role management
  - Firestore for real-time database operations
  - Cloud Functions for backend APIs
  - Cloud Storage for resume and interview media storage
- **AI and NLP Libraries**
  - spaCy, NLTK, Hugging Face Transformers for resume parsing and text analysis
  - TensorFlow or PyTorch for model training and inference
- **Frontend Framework**
  - Flutter SDK for cross-platform mobile and web application development

2. **Software and Platform Dependencies**

- **Firebase Services**
  - Firebase Authentication for user login and role management
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  - Cloud Functions for backend APIs
  - Cloud Storage for resume and interview media storage
- **AI and NLP Libraries**
  - spaCy, NLTK, Hugging Face Transformers for resume parsing and text analysis
  - TensorFlow or PyTorch for model training and inference
- **Frontend Framework**
  - Flutter SDK for cross-platform mobile and web application development

3. **Third-Party and External System Dependencies**
   - Optional integration with **job portals** (e.g., LinkedIn, Rozee.pk) depends on the availability of APIs and access permissions.
   - Email and notification services depend on third-party messaging providers.
   - Cloud hosting services (AWS/GCP) are required for AI model deployment and scalability.

4. **Hardware and Device Dependencies**
   - Camera and microphone availability on user devices for AI mock interviews.
   - Sufficient device processing power to support media capture and real-time communication.
   - Server-side computational resources to execute AI inference within time limits.

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   - Sufficient device processing power to support media capture and real-time communication.
   - Server-side computational resources to execute AI inference within time limits.

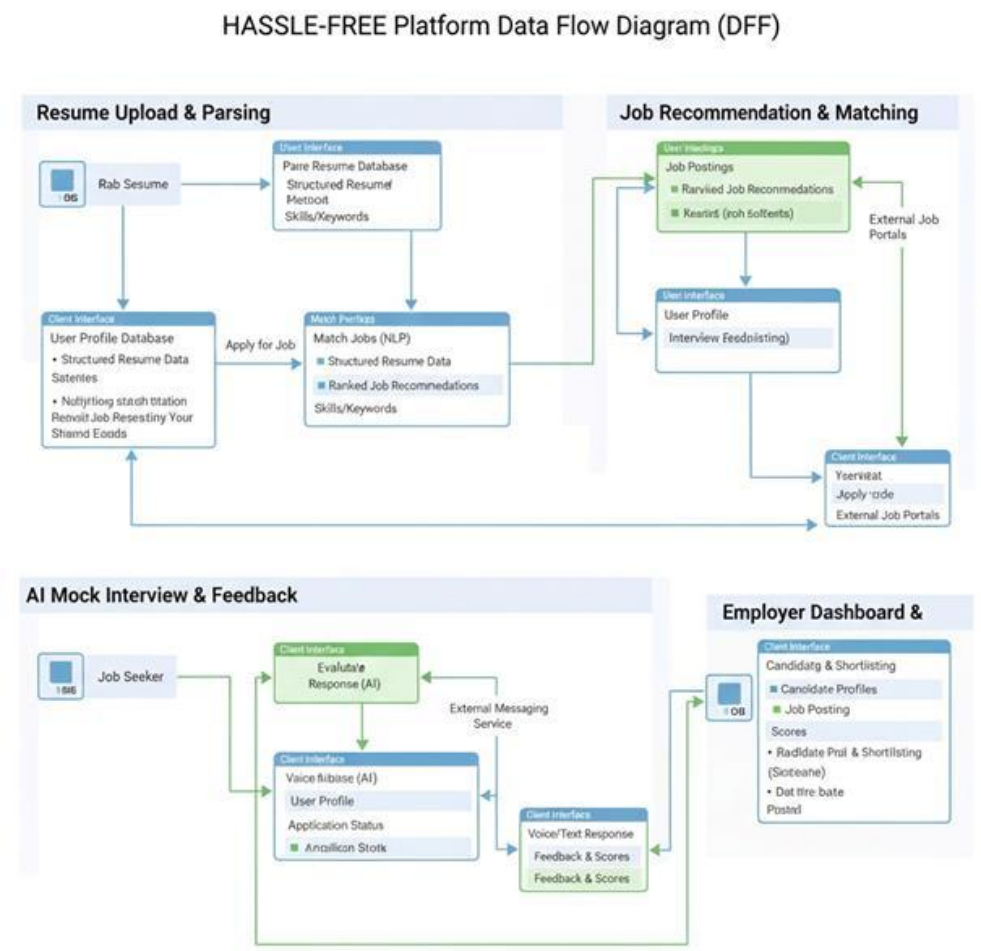
- **Impact of Assumption or Dependency Changes**

If any assumption proves invalid or any dependency becomes unavailable or changes significantly:

- AI accuracy and system performance may degrade.
- Certain features (e.g., mock interviews, automated job applications) may become partially or fully unavailable.
- System design or implementation may require modification.
- Project timelines and deliverables may be affected.

3. Technical Architecture

This section describes the **technical architecture** of the HASSLE-FREE system. It explicitly addresses the architectural questions defined in the SDS Phase-II template and presents a structured explanation of the system's components, technologies, deployment environment, and interaction mechanisms.



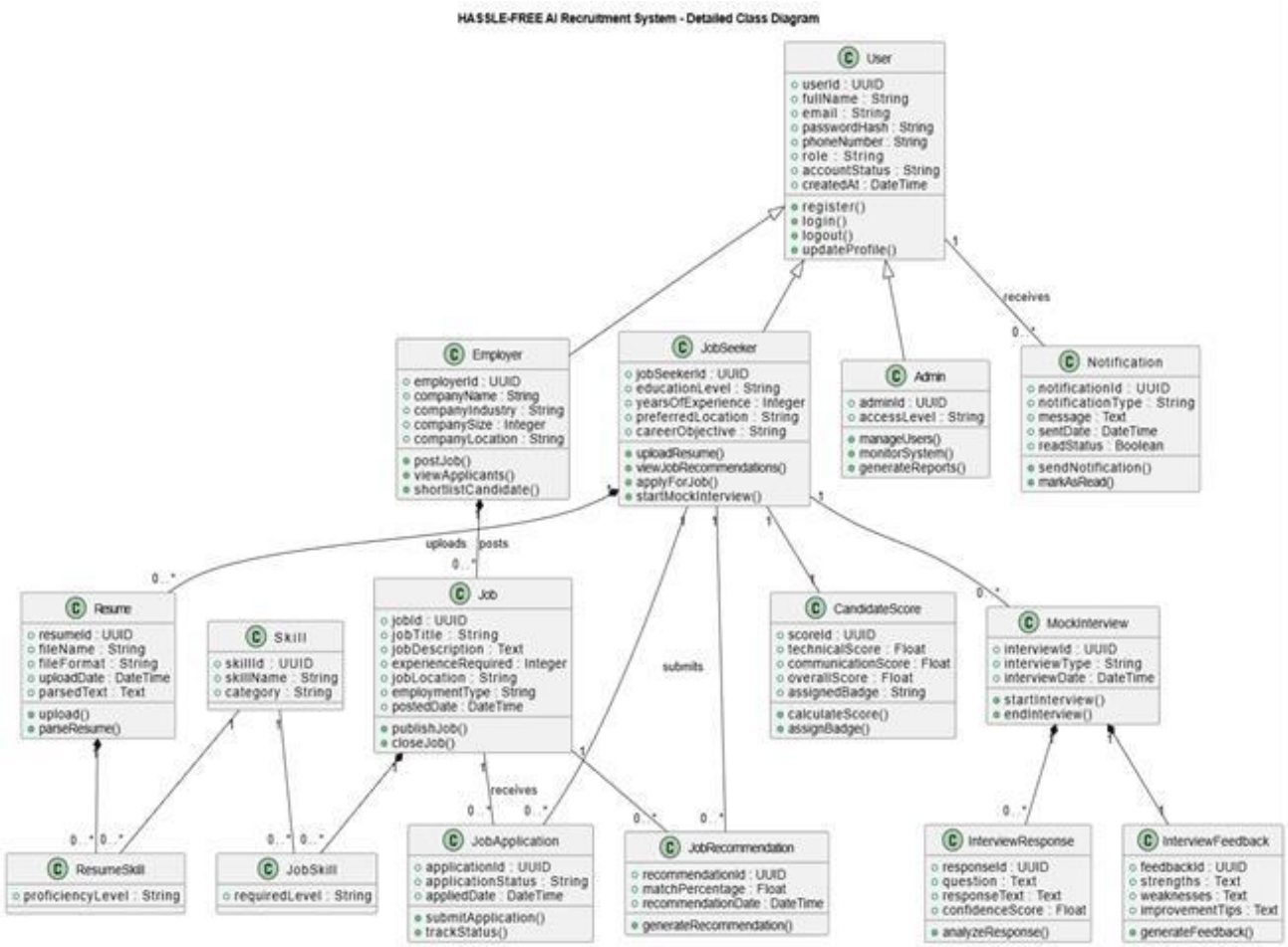
3.1 Application and Data Architecture

This section describes the **application components, processing units, and data organization** of the HASSLE-FREE system. The architecture is designed to support intelligent recruitment, career enhancement, and scalable data management. A combination of **logical and physical models** is used to represent the system.

The following diagrams are used to explain the application and data architecture in detail:

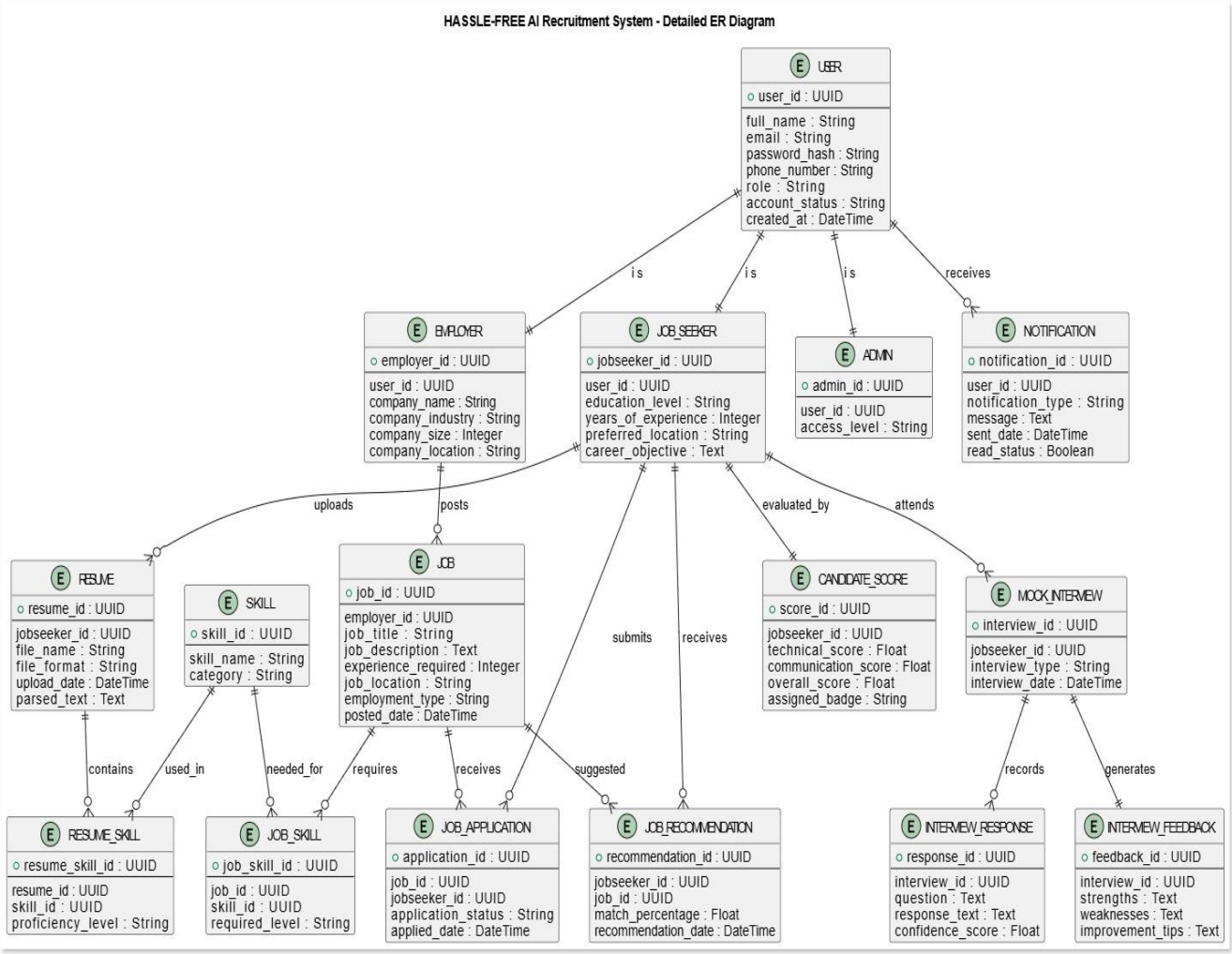
3.1.1 Class Diagram

The class diagram represents the **static structure** of the system and defines the main classes, their attributes, methods, and relationships.
Key classes include User, JobSeeker, Employer, Resume, Job, Application, Interview, and AIEngine. Inheritance is used to model different user roles, while associations and compositions represent relationships between resumes, jobs, applications, and interviews. This diagram provides a complete object-oriented view of the system design.



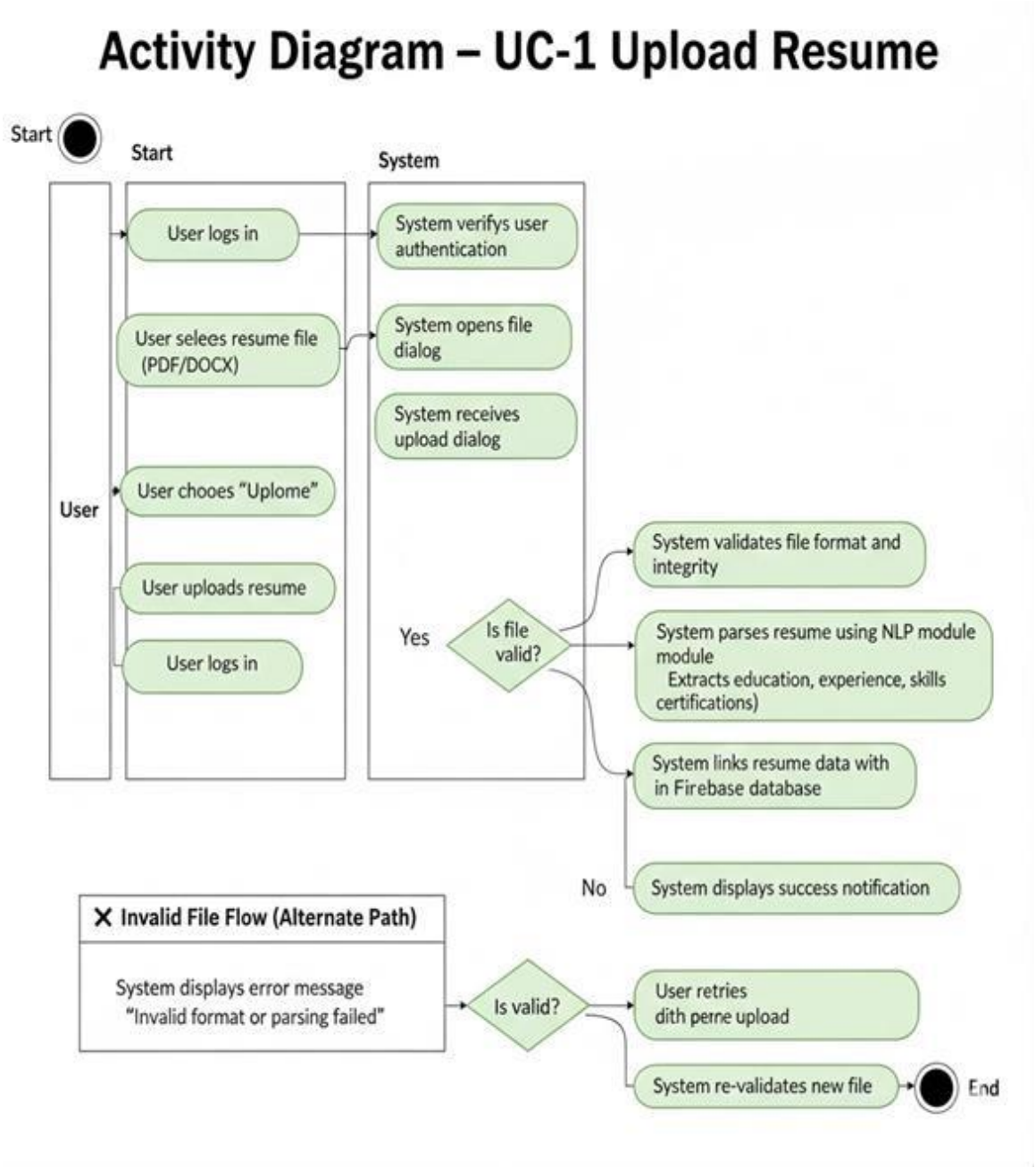
3.1.2 Entity Relationship (ER) Diagram

The ER diagram describes the **data model** and illustrates how information is stored in the database. It identifies entities such as *User*, *Resume*, *Job*, *Application*, and *Interview*, along with their primary keys and relationships.
This diagram ensures data normalization, consistency, and integrity within the Firestore database.

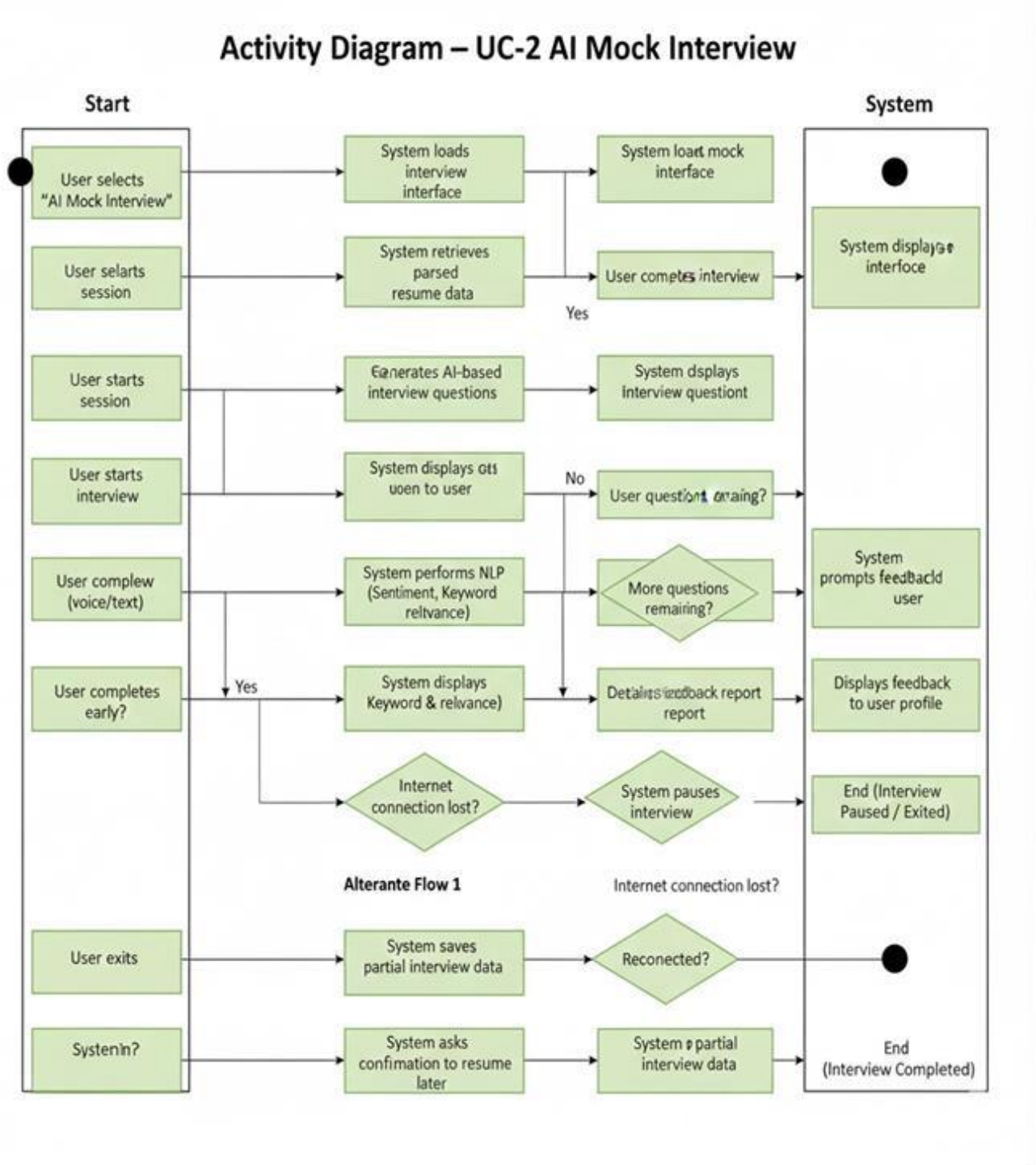


3.1.3 ACTIVITY DIAGRAM:

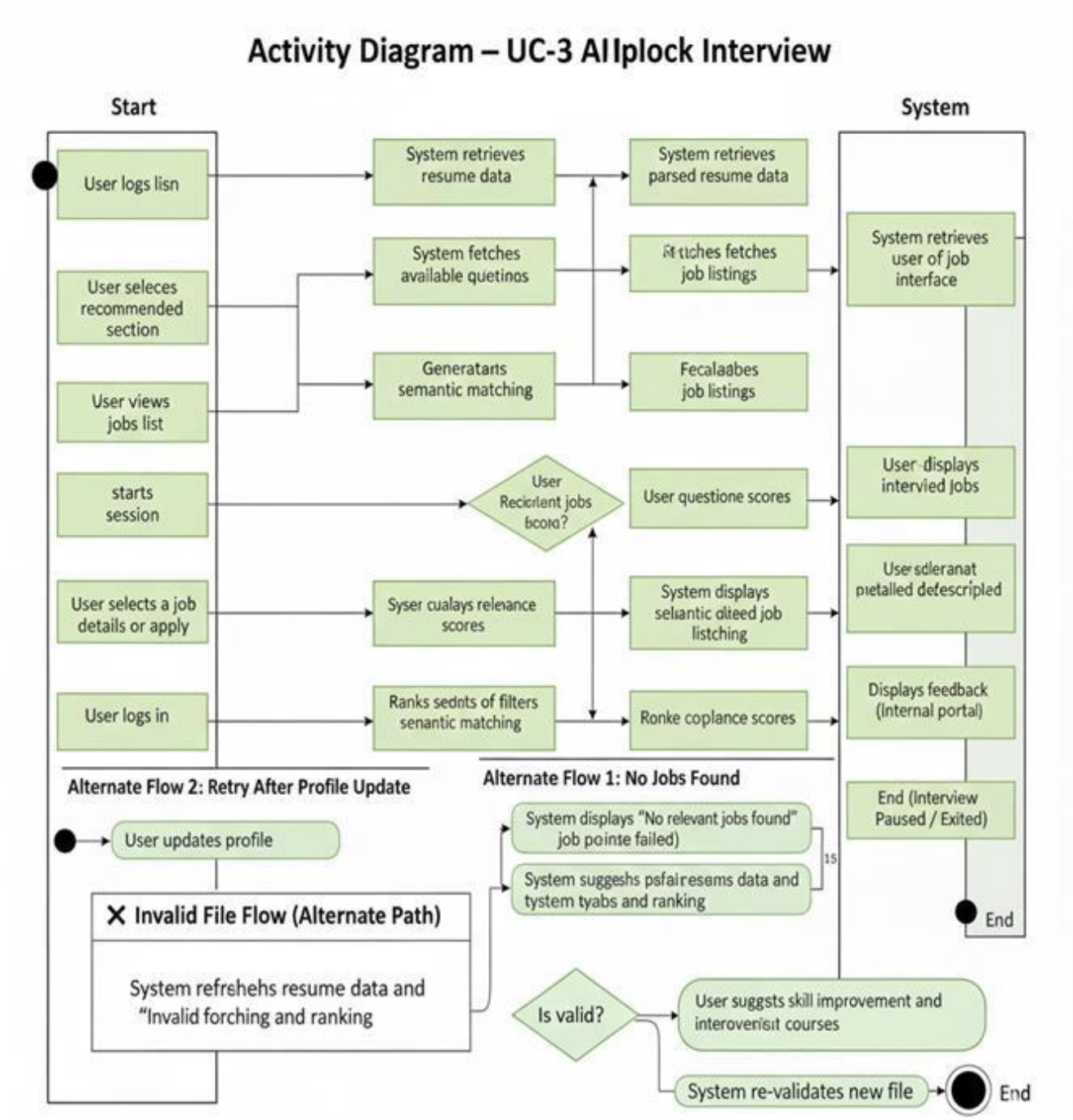
→Activity 1 For Upload Resume:



→Activity 2 For AI Mock Interview:



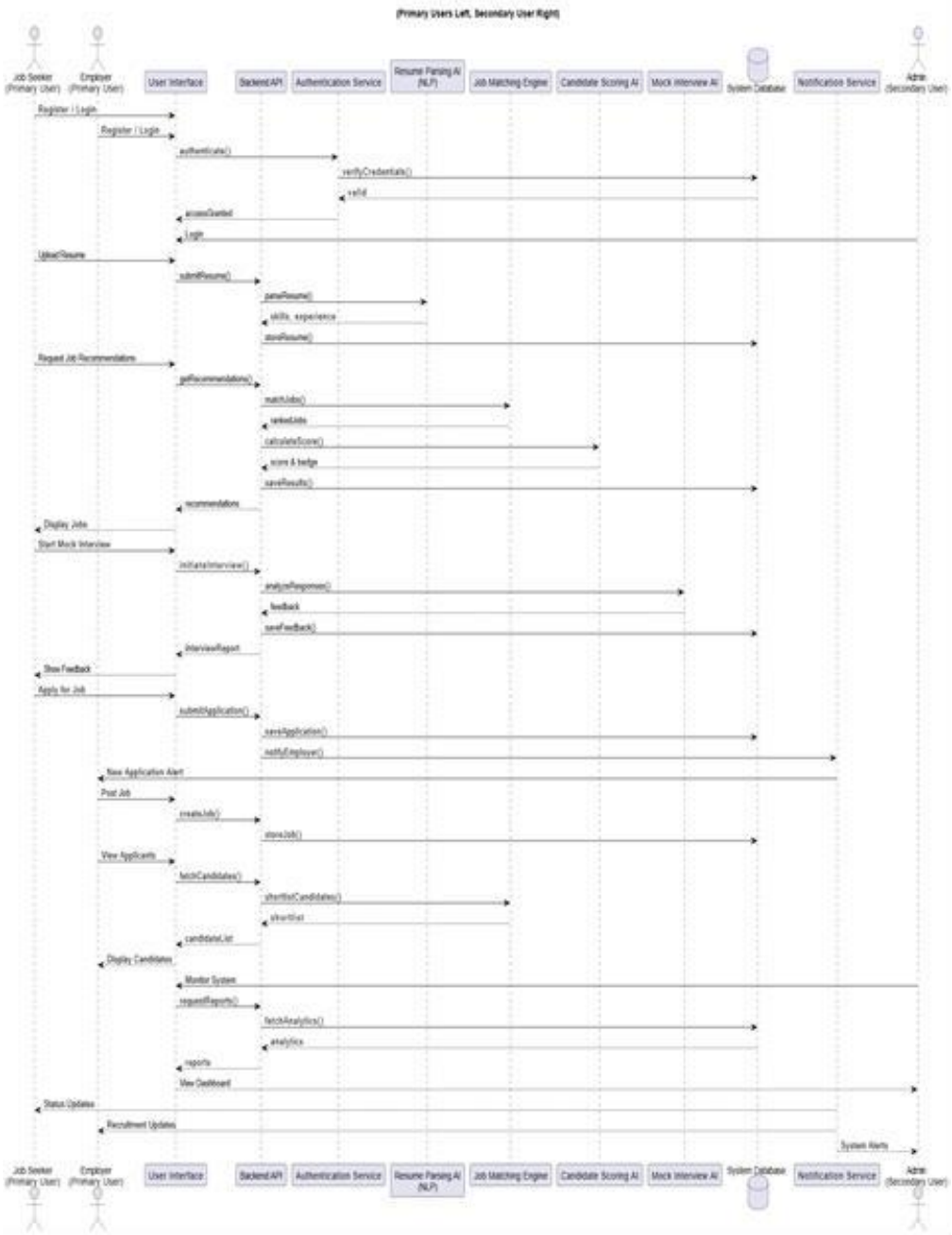
→Activity 3 All Activity Diagram:



3.2 Component Interactions and Collaborations

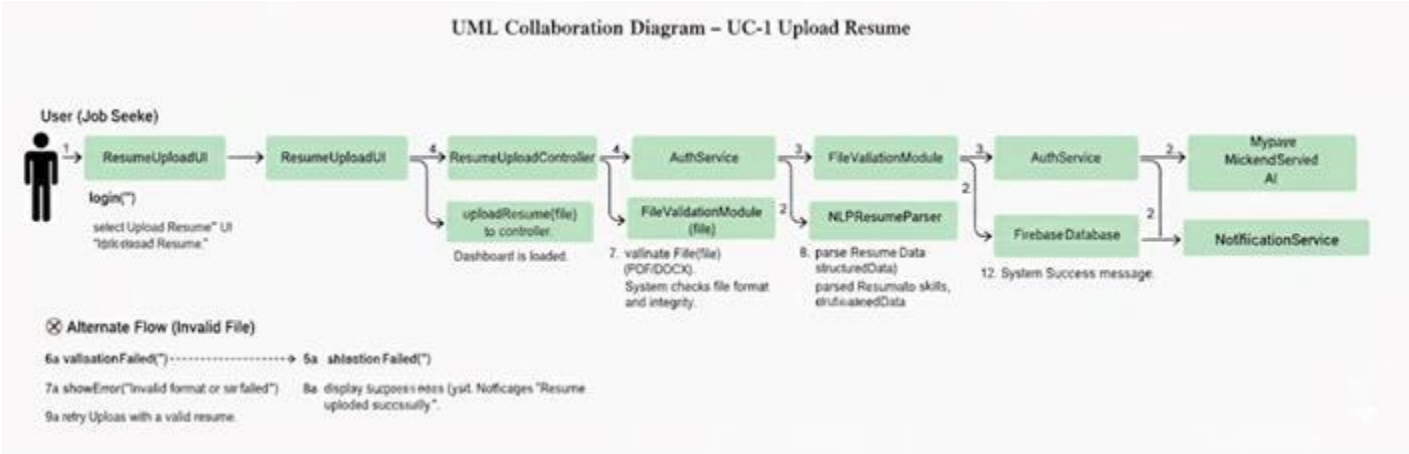
The component diagram presents a **logical decomposition** of the system into major functional components such as the User Interface, Authentication Service, Application Services, AI Processing Services, and Data Management Services.
It shows how these components interact through defined interfaces and APIs.

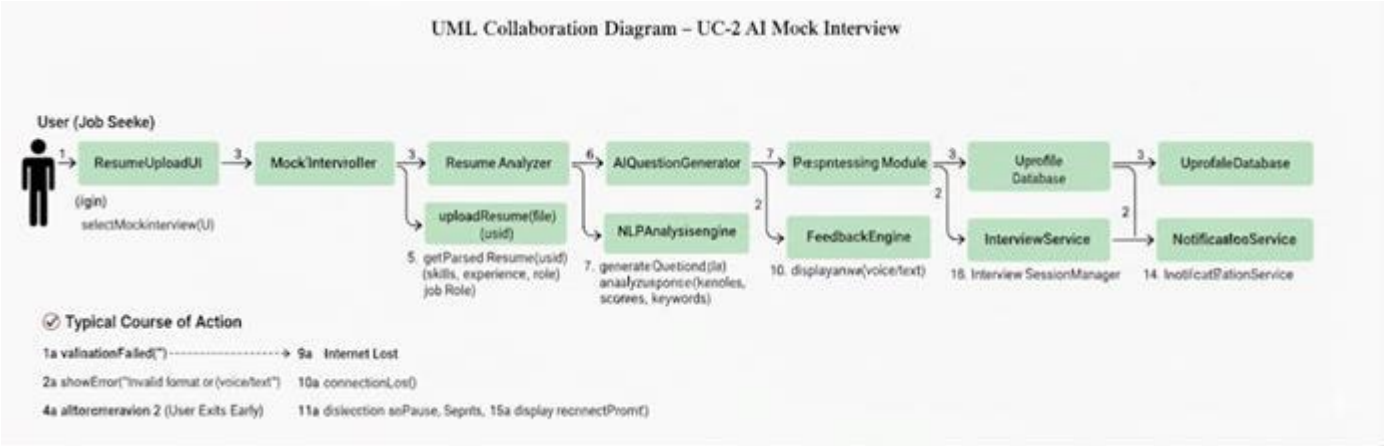
SEQUENCE DIAGRAM:

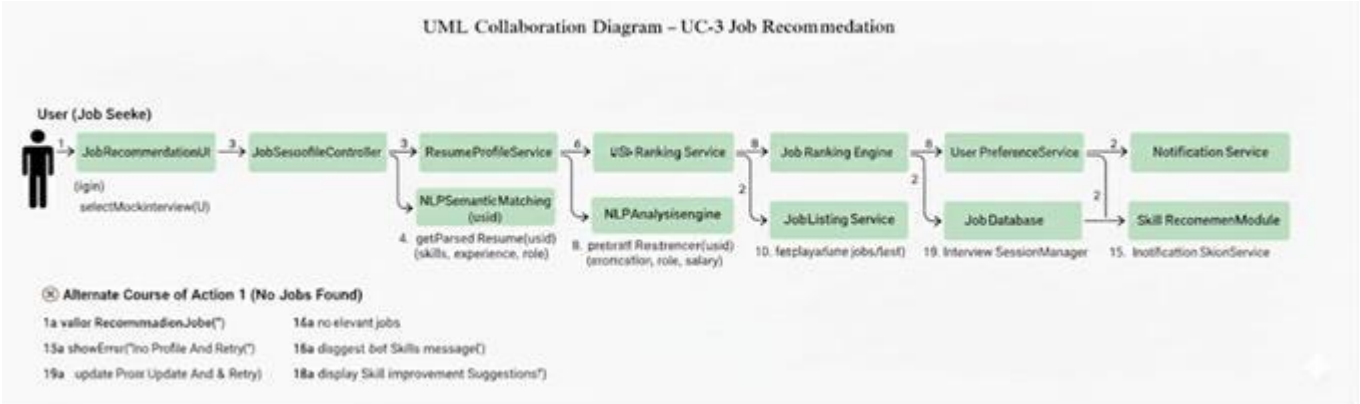


COLLABORATION DIAGRAM:

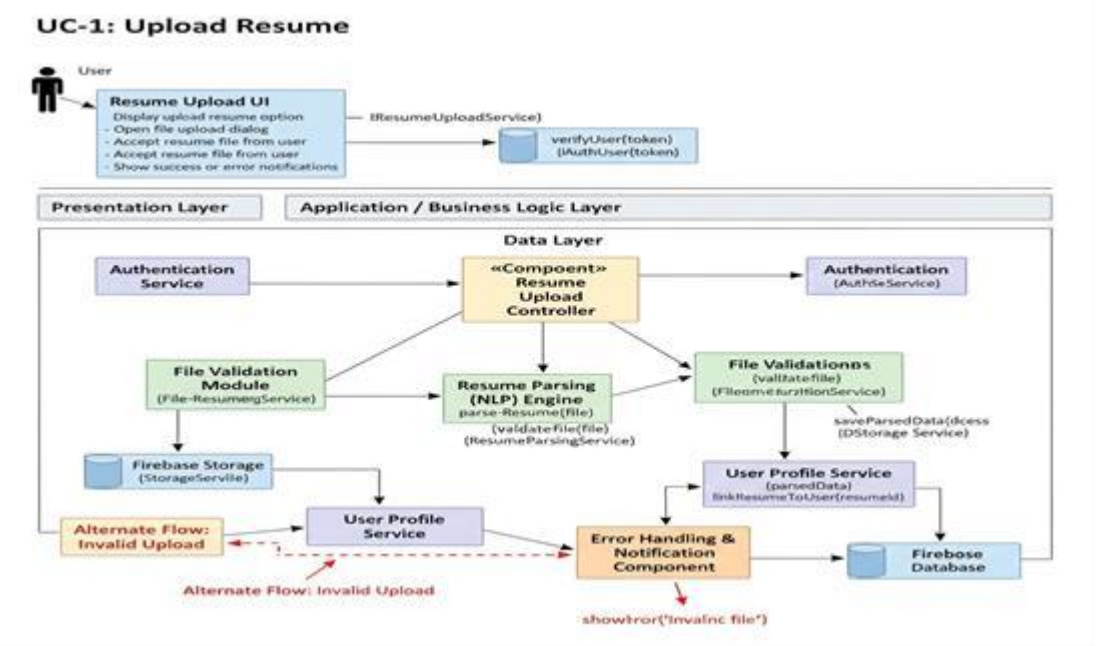
USER FLOW:



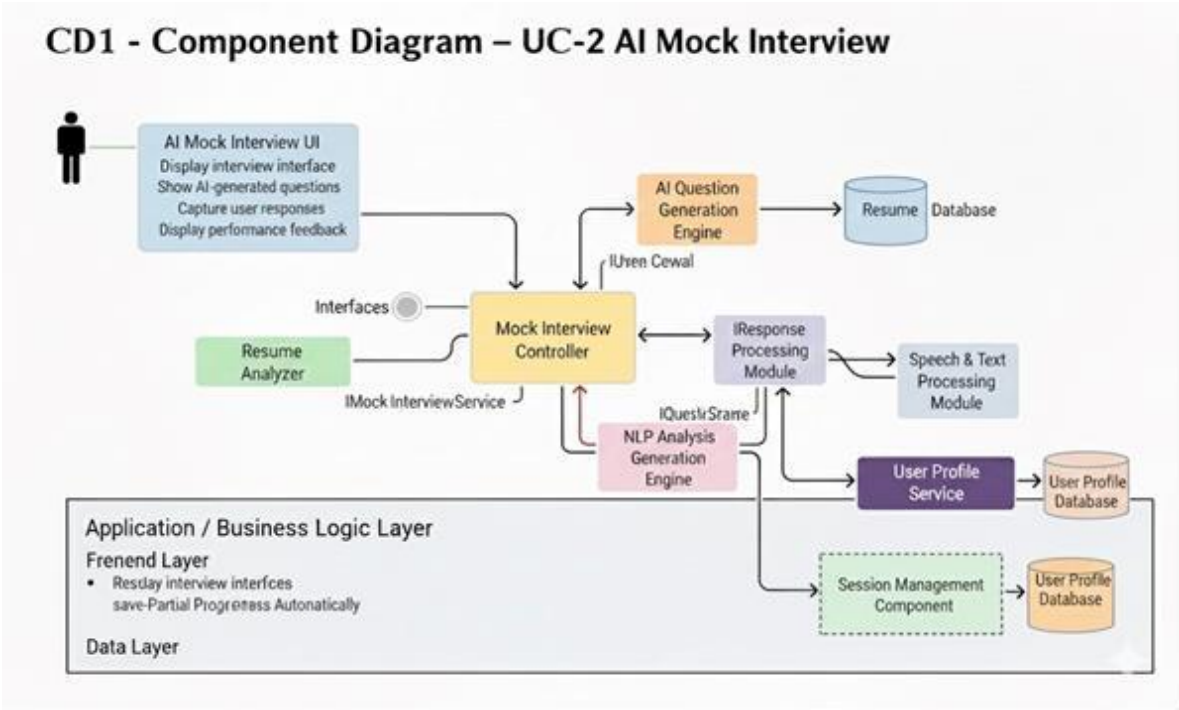




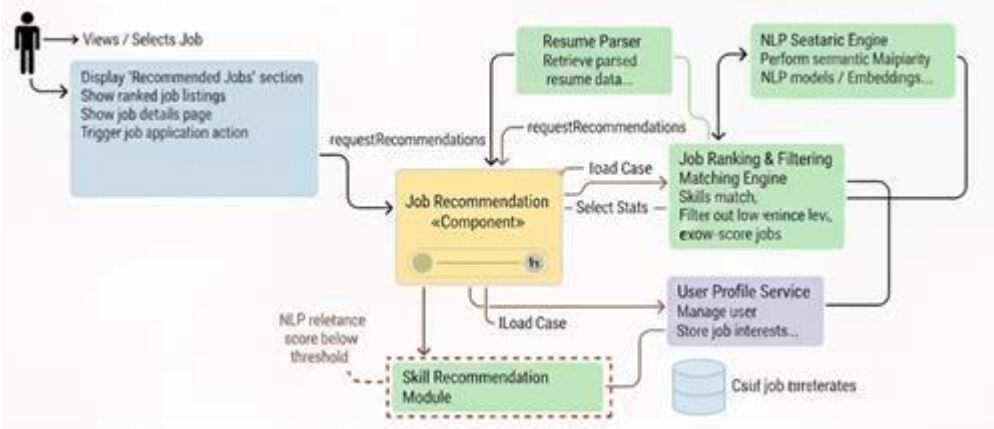
CD1-Upload Resume (UC-1):



CD2-AI-Mock Interview (UC-2):

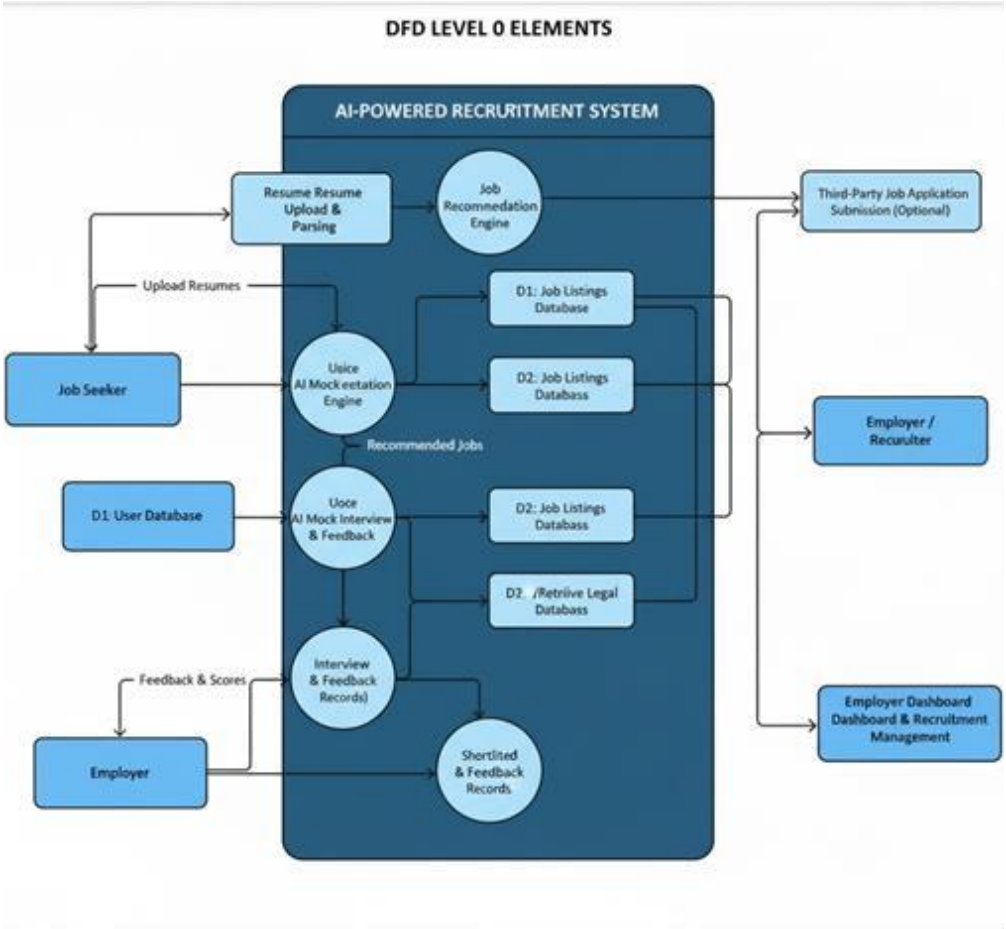


CD3-For searching job (UC-3):

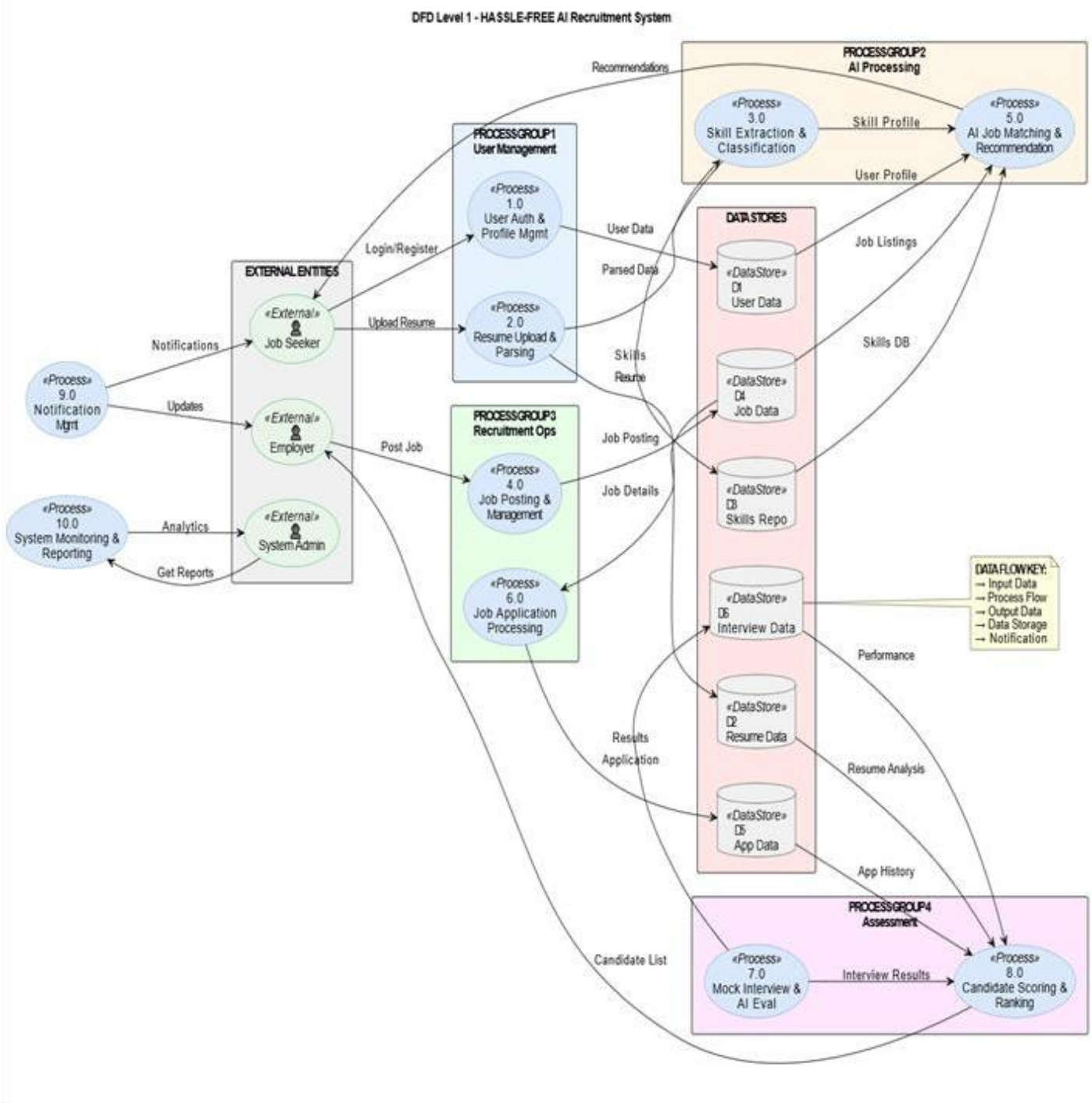


- **DATA FLOW DIAGRAM:**

DFD LEVEL 0:



DFD LEVEL-1:


DFD Level 1 - HASSLE-FREE AI Recruitment System
PROCESSGROUP2
AI Processing
«Process»
3.0
Skill Extraction &
Classification
Skill Profile
«Process»
5.0
AI Job Matching &
Recommendation
User Profile
Recommendations
PROCESSGROUP1
User Management
«Process»
1.0
User Auth &
Profile Mgmt
«Process»
2.0
Resume Upload &
Parsing
User Data
Parsed Data
Login/Register
Upload Resume
EXTERNAL ENTITIES
«External»
Job Seeker
«External»
Employer
«External»
System Admin
Notifications
Updates
Analytics
Get Reports
«Process»
9.0
Notification
Mgmt
«Process»
10.0
System Monitoring &
Reporting
DATA STORES
«DataStore»
D1
User Data
«DataStore»
D4
Job Data
«DataStore»
D3
Skills Repo
«DataStore»
D6
Interview Data
«DataStore»
D2
Resume Data
«DataStore»
D5
App Data
Job Listings
Skills DB
Skills
Resume
Job Posting
Job Details
Post Job
PROCESSGROUP3
Recruitment Ops
«Process»
4.0
Job Posting &
Management
«Process»
6.0
Job Application
Processing
DATA FLOW KEY:
→ Input Data
→ Process Flow
→ Output Data
→ Data Storage
→ Notification
Performance
Results
Application
Resume Analysis
App History
PROCESSGROUP4
Assessment
«Process»
7.0
Mock Interview &
AI Eval
Interview Results
«Process»
8.0
Candidate Scoring &
Ranking
Candidate List

3.3 Design Reuse and Design Patterns

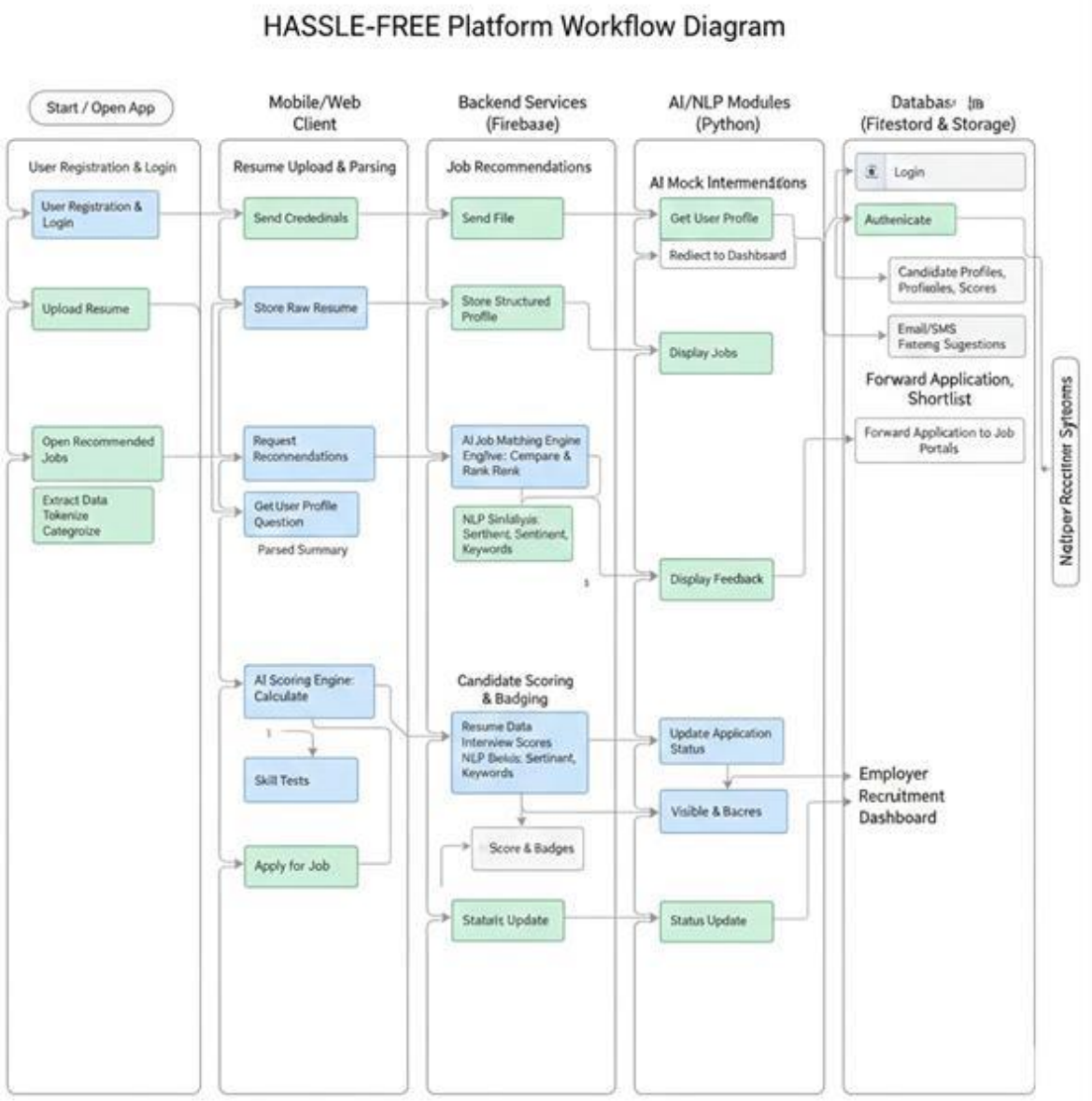
The HASSLE-FREE system employs strategic design reuse through established architectural patterns to ensure scalability and maintainability. The system adopts **a** microservices architecture for AI modules (resume parsing, job matching, interview analysis), allowing independent deployment and scaling. We implement the Repository Pattern for all Firebase Firestore operations and use reusable authentication, file upload, and notification components. Key design patterns include Factory Pattern for interview question generation**,** Strategy Pattern for multiple job matching algorithms**,** Observer Pattern for real-time notifications, and Adapter Pattern for external job portal integration, creating a robust, modular foundation.

3.4 Technology Architecture

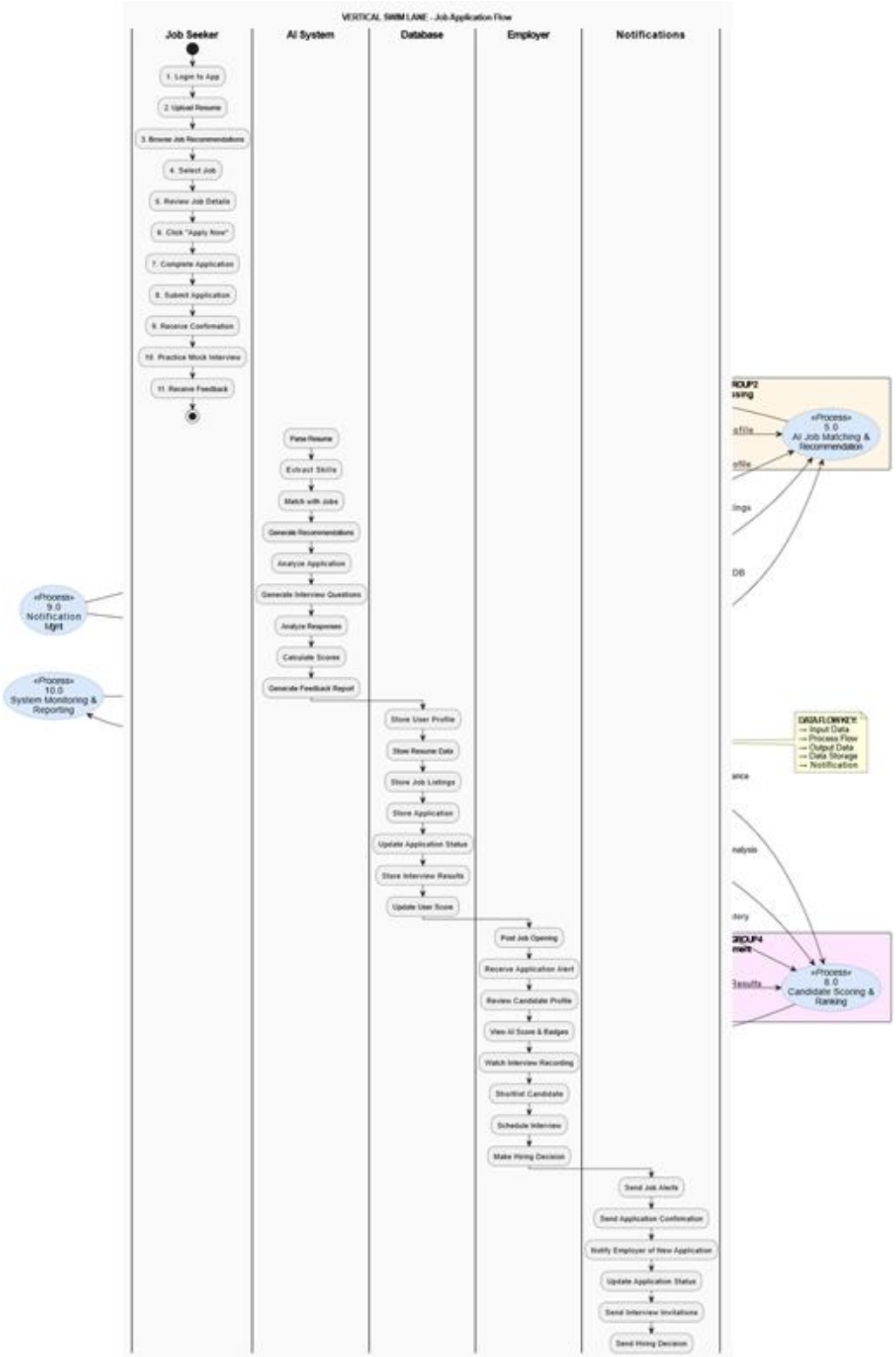
The system leverages a **cloud-native, multi-tier infrastructure** with Flutter mobile/web apps hosted on **Firebase Hosting**, backend logic via **Firebase Cloud Functions**, and real-time data in **Firebase Firestore**. AI services run on **AWS Lambda** as containerized microservices, while **Firebase Cloud Storage** handles resume and interview files. Connectivity requires stable internet with HTTPS/TLS 1.2+ encryption and RESTful APIs, supporting both **online mode** for full functionality and **limited offline mode** for cached content. The architecture features **horizontal auto-scaling** for AI services, global CDN distribution, and cloud-native load balancing to ensure performance under varying loads.

4. Screenshots/Prototype

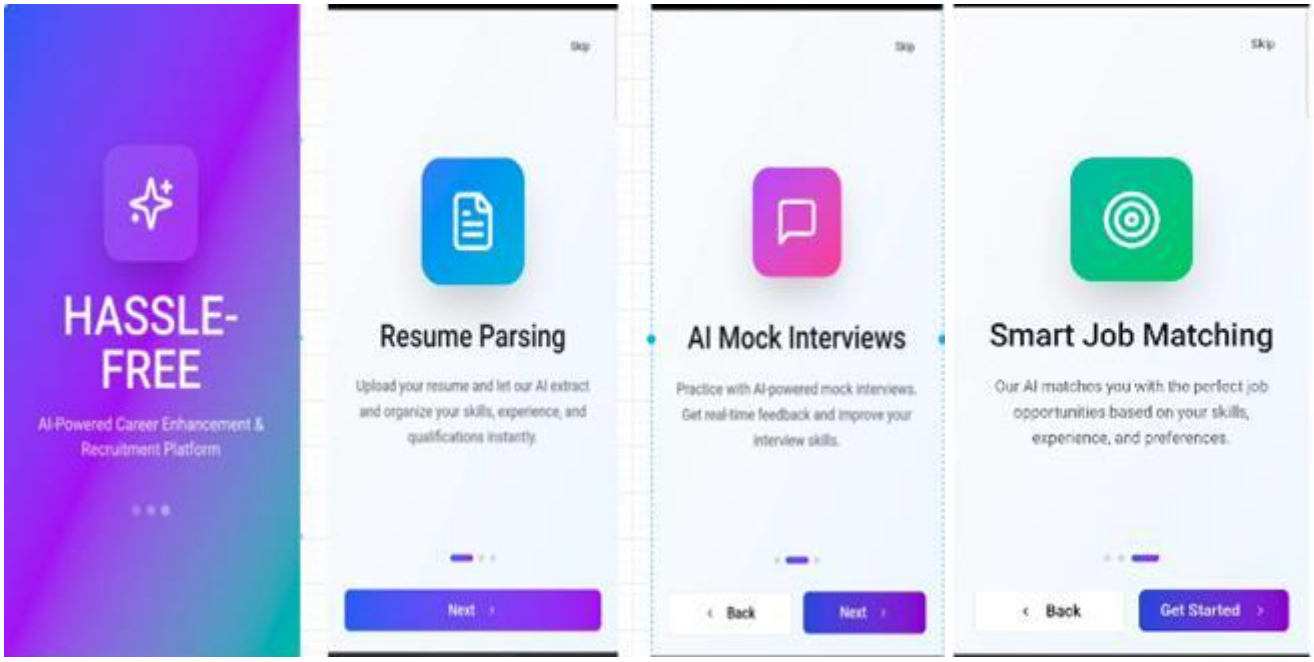
4.1 Workflow

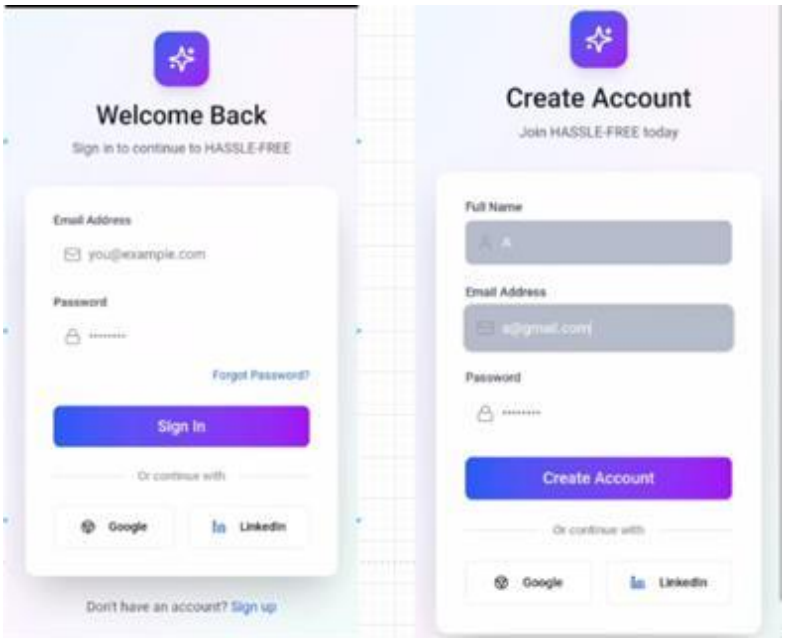


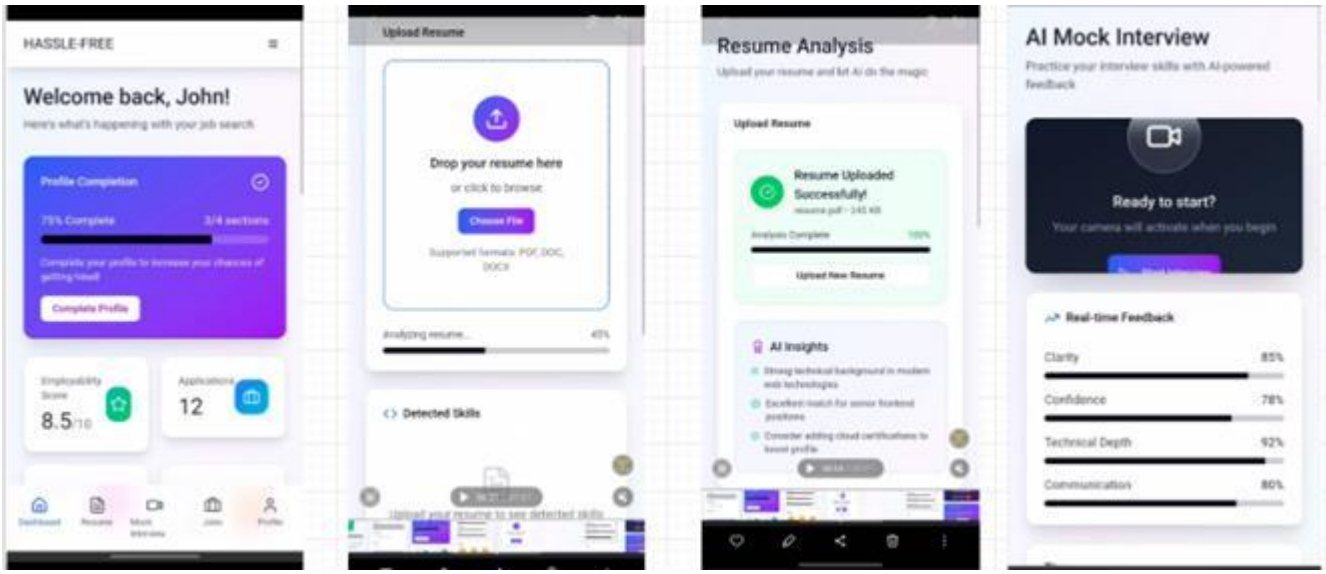
SWIM-LANE DIAGRAM:

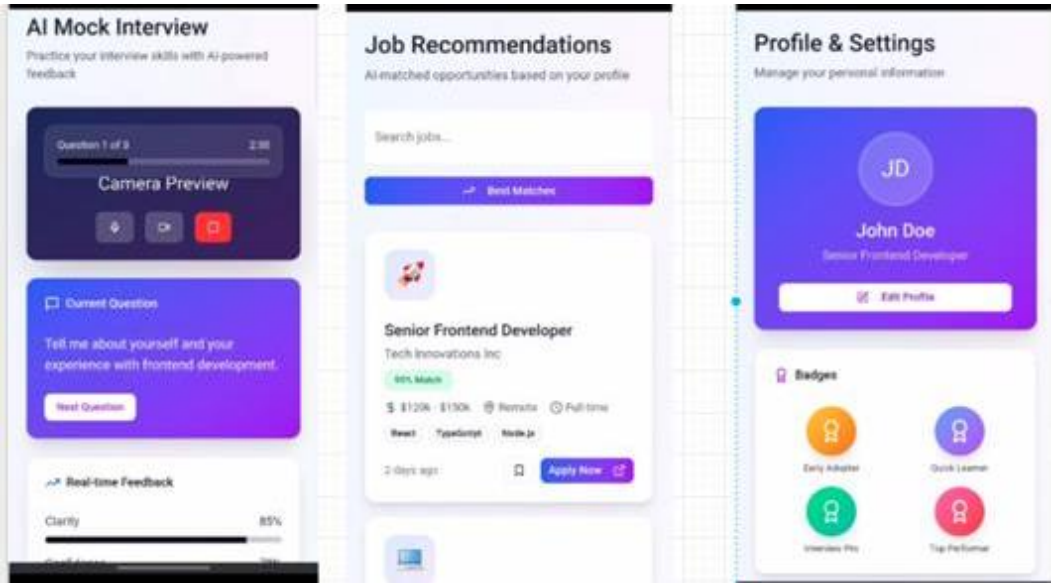


4.2 Screens

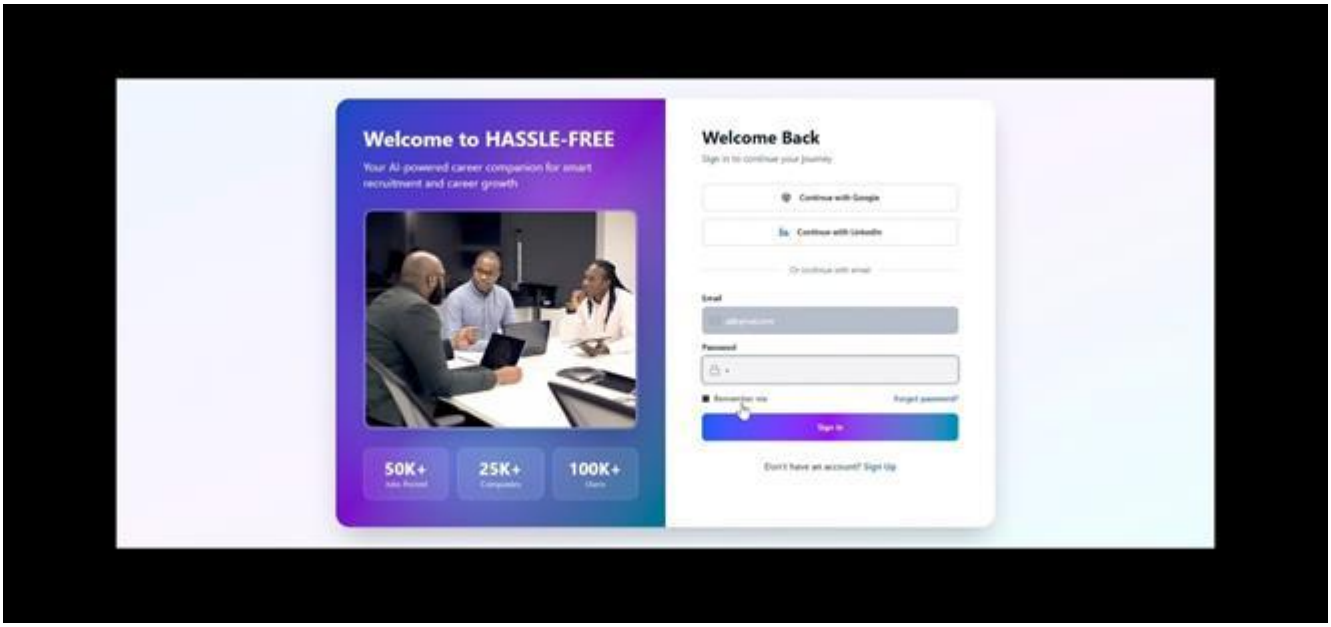

HASSLE-FREE
AI-Powered Career Enhancement & Recruitment Platform
Skip
Resume Parsing
Upload your resume and let our AI extract and organize your skills, experience, and qualifications instantly.
Next
Skip
AI Mock Interviews
Practice with AI-powered mock interviews. Get real-time feedback and improve your interview skills.
Back
Next
Skip
Smart Job Matching
Our AI matches you with the perfect job opportunities based on your skills, experience, and preferences.
Back
Get Started

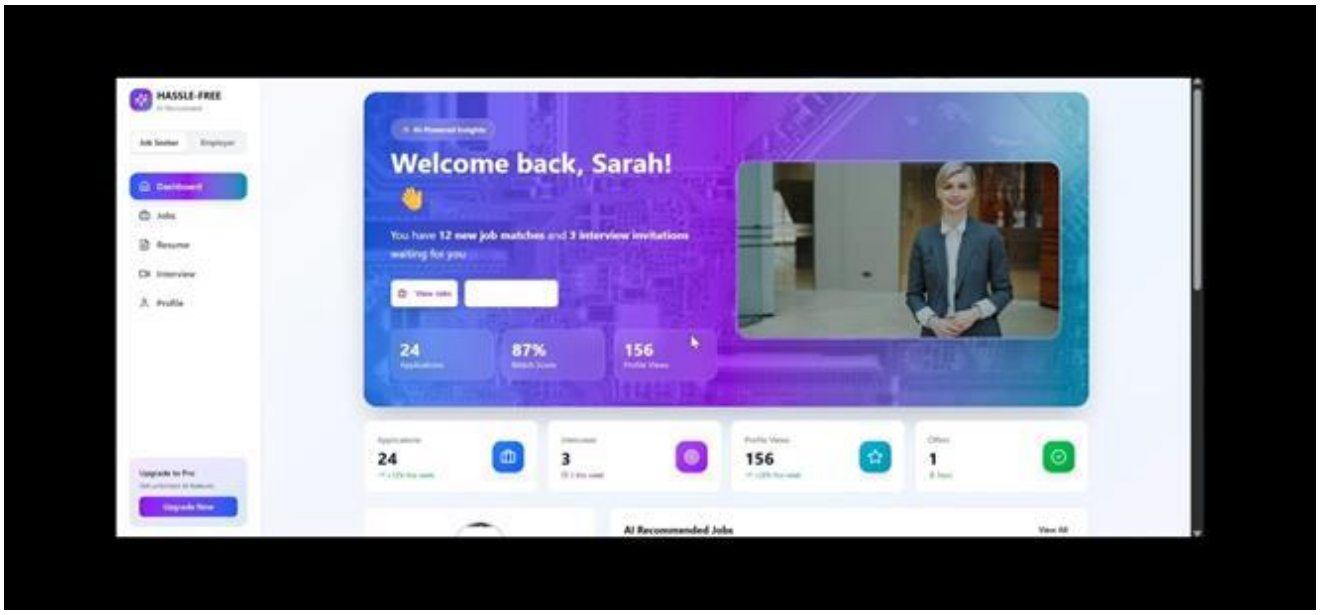

Welcome Back
Sign in to continue to HASSLE-FREE
Email Address
you@example.com
Password
Forgot Password?
Sign In
Or continue with
Google
LinkedIn
Don't have an account? Sign up
Create Account
Join HASSLE-FREE today
Full Name
Email Address
Password
Create Account
Or continue with
Google
LinkedIn

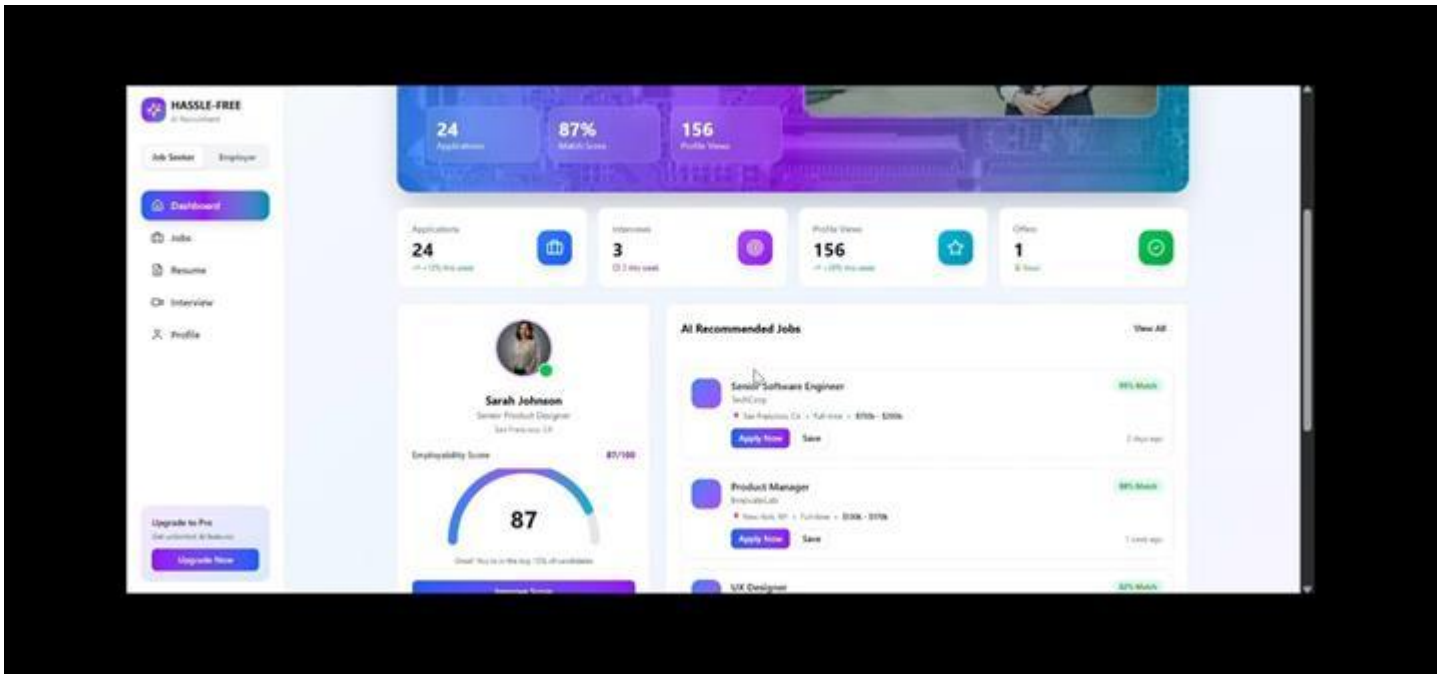

HASSLE-FREE
Welcome back, John!
Profile Completion
Complete Profile
8.5
12
Upload Resume
Drop your resume here
or click to browse
Choose File
Detected Skills
Resume Analysis
Upload Resume
Resume Uploaded Successfully!
Upload New Resume
AI Insights
AI Mock Interview
Ready to start?
Real-time Feedback
Clarity
Confidence
Technical Depth
Communication

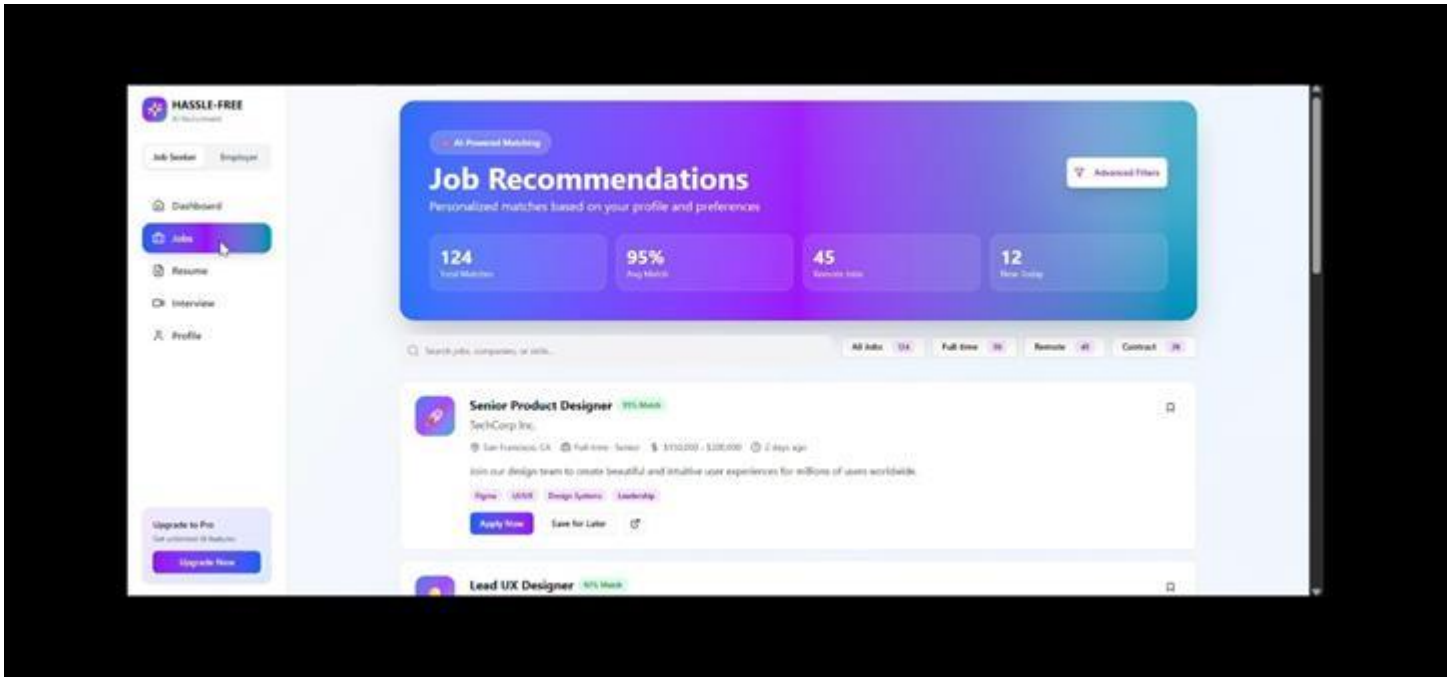

AI Mock Interview
Camera Preview
Current Question
Tell me about yourself and your experience with frontend development.
Next Question
Real-time Feedback
Clarity
Job Recommendations
AI-matched opportunities based on your profile
Search jobs...
Best Matches
Senior Frontend Developer
Tech Innovations Inc
Apply Now
Profile & Settings
Manage your personal information
JD
John Doe
Edit Profile
Badges

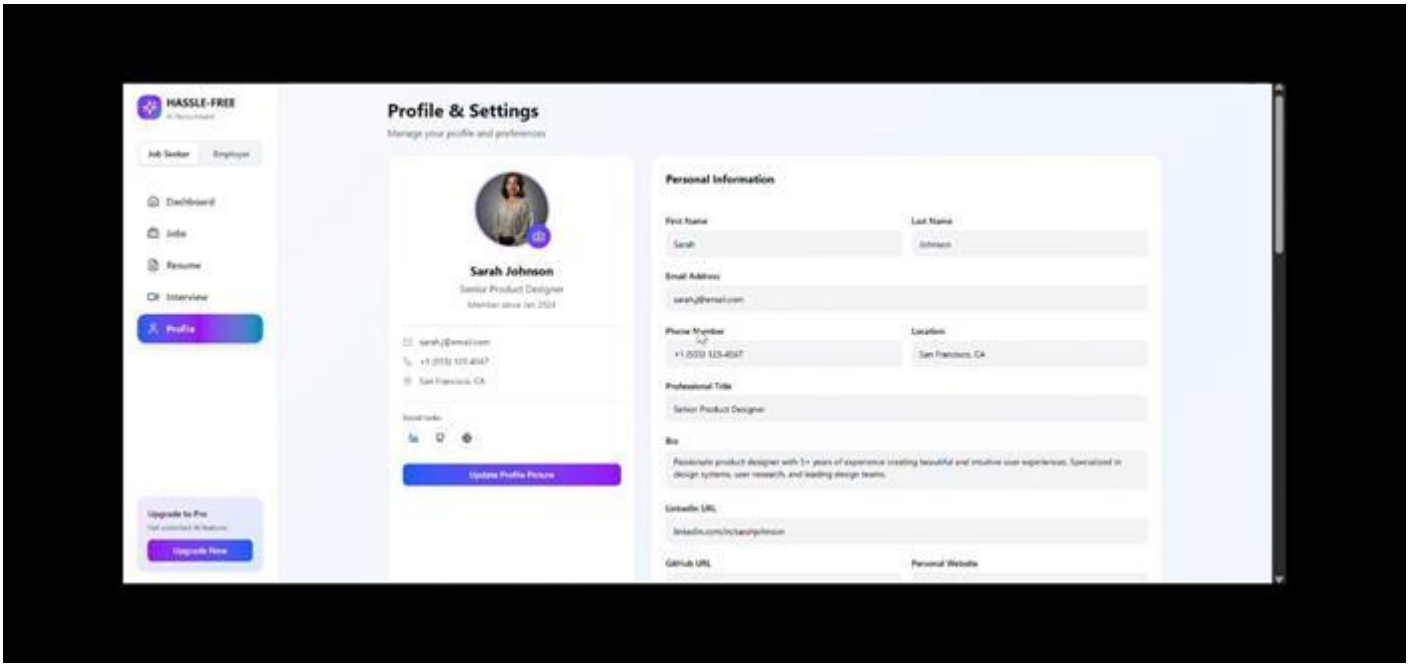
➔ WEB DASHBOARD:

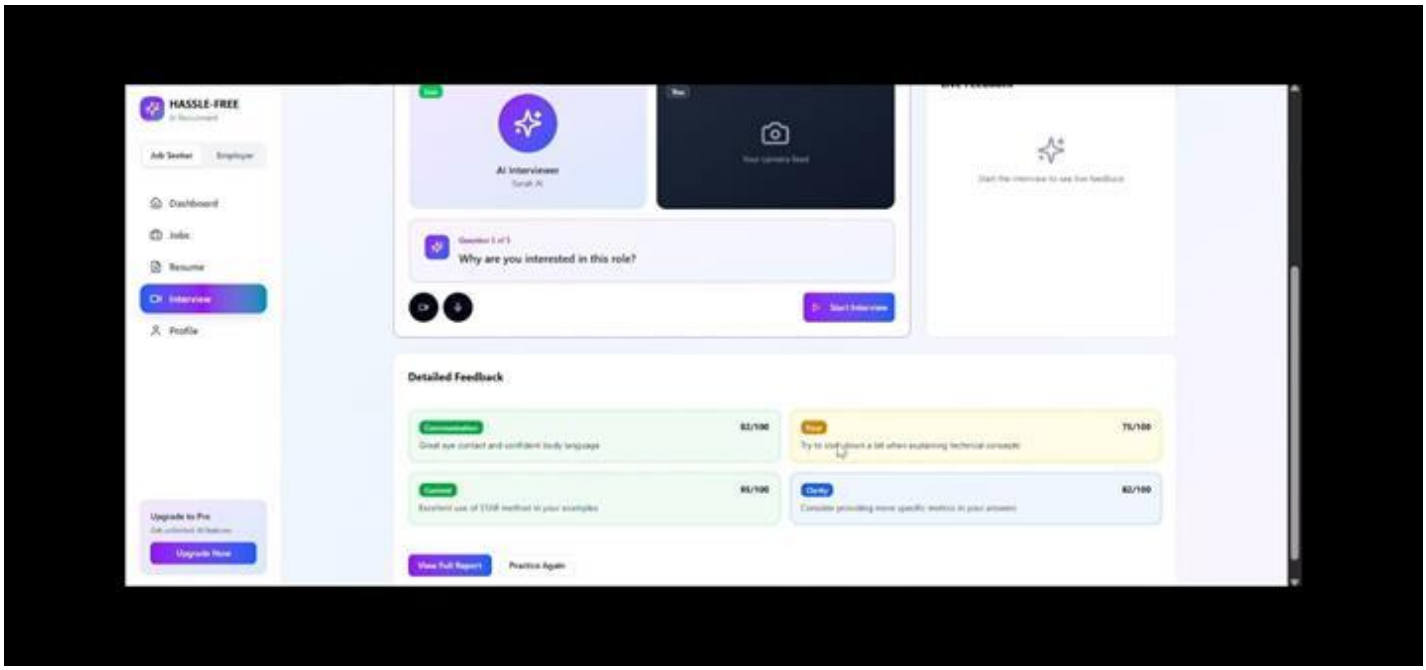


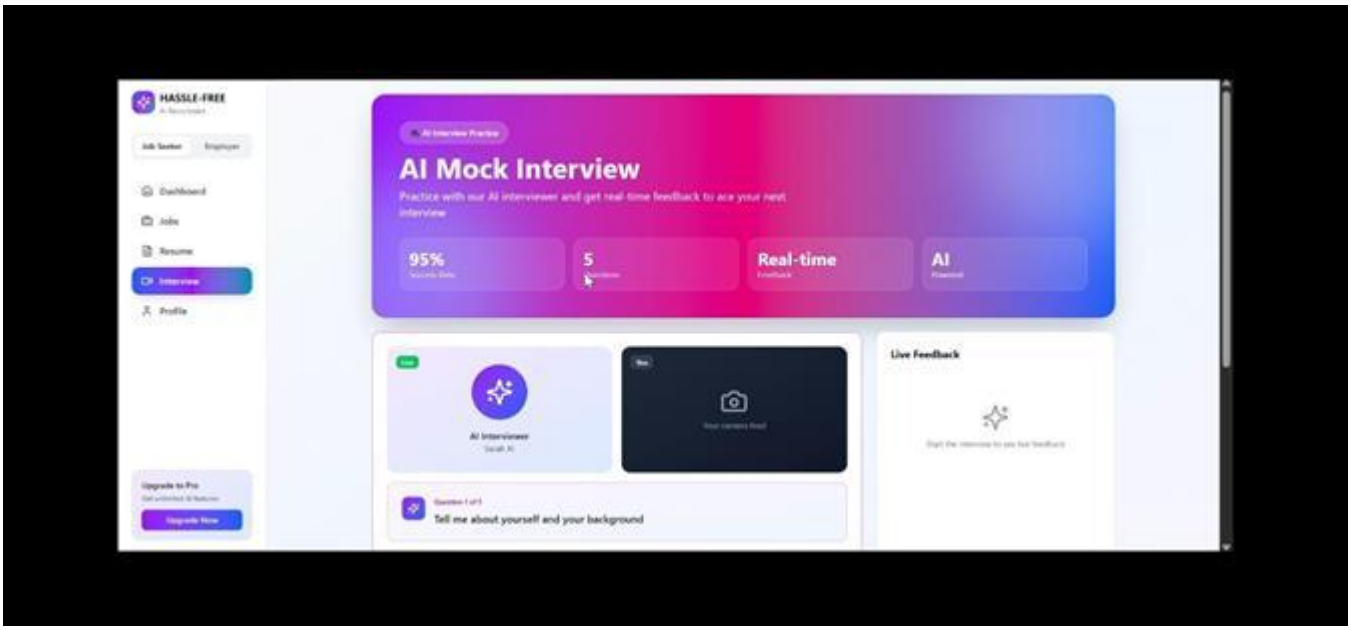


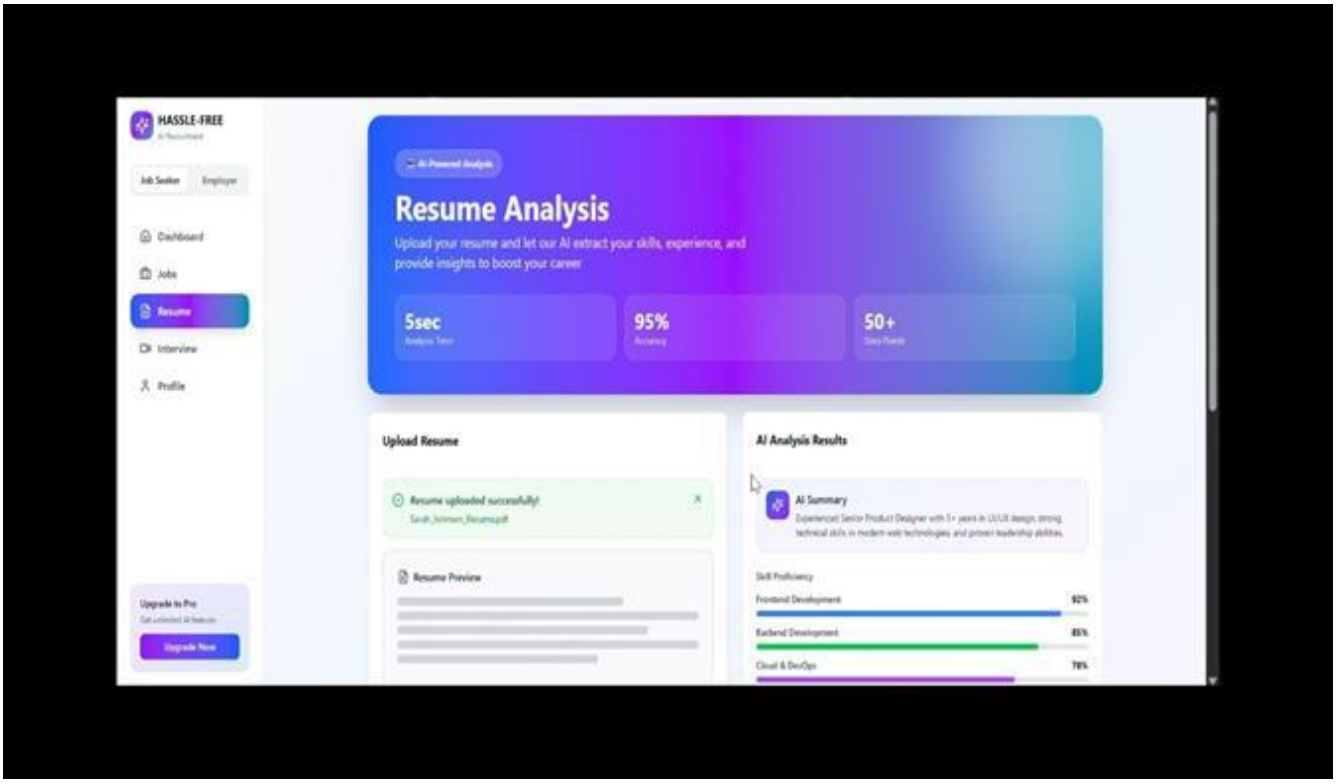

HASSLE-FREE
Job Seeker
Employer
Dashboard
Jobs
Resume
Interview
Profile
Upgrade to Pro
Upgrade Now
24
87%
156
24
3
156
1
Sarah Johnson
Employability Score
87
AI Recommended Jobs
View All
Senior Software Engineer
Apply Now
Save
Product Manager
Apply Now
Save
UX Designer


HASSLE-FREE
Job Seeker
Employer
Dashboard
Jobs
Resume
Interview
Profile
Upgrade to Pro
Upgrade Now
AI Powered Matching
Job Recommendations
Personalized matches based on your profile and preferences
Advanced Filters
124
95%
45
12
All Jobs
Full-time
Remote
Contract
Senior Product Designer
SecCorp Inc.
Apply Now
Save for Later
Lead UX Designer


HASSLE-FREE
Job Seeker
Employer
Dashboard
Jobs
Resume
Interview
Profile
Upgrade to Pro
Upgrade Now
Profile & Settings
Manage your profile and preferences
Sarah Johnson
Senior Product Designer
Update Profile Picture
Personal Information
First Name
Sarah
Last Name
Johnson
Email Address
Phone Number
Location
San Francisco, CA
Professional Title
Senior Product Designer
Bio
LinkedIn URL
GitHub URL
Personal Website


HASSLE-FREE
Job Seeker
Employer
Dashboard
Jobs
Resume
Interview
Profile
Upgrade to Pro
Upgrade Now
AI Interviewer
Why are you interested in this role?
Start Interview
Detailed Feedback
82/100
75/100
85/100
82/100
View Full Report
Practice Again


HASSLE-FREE
Dashboard
Jobs
Resume
Interview
Profile
AI Mock Interview
95%
5
Real-time
AI
Live Feedback
AI Interviewer
Tell me about yourself and your background


HASSLE-FREE
Job Seeker
Employer
Dashboard
Jobs
Resume
Interview
Profile
Resume Analysis
Upload your resume and let our AI extract your skills, experience, and provide insights to boost your career
5sec
95%
50+
Upload Resume
Resume uploaded successfully!
Resume Preview
AI Analysis Results
AI Summary
Skill Proficiency
Frontend Development
Backend Development
Cloud & DevOps

4.3 Additional Information

The HASSLE-FREE platform requires the following development setup:

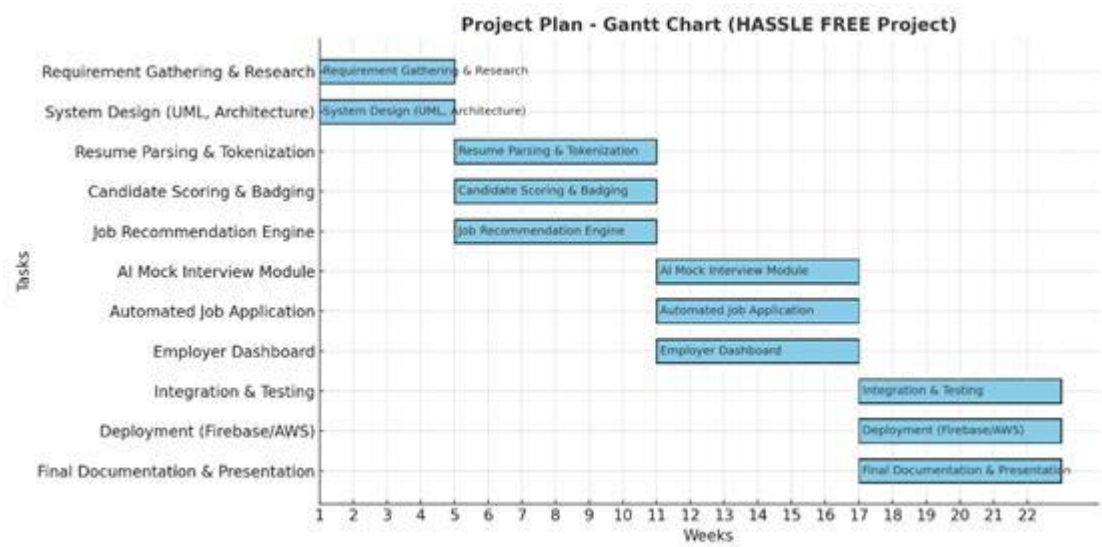
Development Tools Stack:

- **IDE**: Visual Studio Code with Flutter/Dart extensions
- **Version Control**: Git with GitHub repository for team collaboration
- **Package Manager**: Pub for Flutter, pip for Python AI services
- **Containerization**: Docker Desktop for local AI service testing
- **Database GUI**: Firebase Console and Firestore browser extension

5. Other Design Details

There is no such other detail that we shared in this section

6. Revised Project Plan



GANTT CHART

Phase 1. Requirement & Research Status:

Fully Completed
Duration: Weeks 1-7 (Sept-Oct 2025)
Deliverables Achieved:

- Complete SRS Document (Version 1.0)
- UML Diagrams (Use Case, Class, Sequence, Activity)
- DFD Level 0 & Level 1 Diagrams
- Market Research & Competitor Analysis
- Technical Architecture Design
- Risk Assessment Document

Phase 2: Modules Development (IN PROGRESS - 65%) Status:

Currently Active
Duration: Weeks 8-16 (Nov 2025 - Jan 2026)
Progress Breakdown:

Sub-module Progress:

1. **Resume Parsing & Tokenization (NLP) - 85% Complete**
    - spaCy model integration
    - PDF/DOCX parsing functionality
    - Skill extraction algorithm
    - Experience classification refinement
2. **Candidate Scoring & Badging Engine - 70% Complete**
    - Scoring algorithm framework
    - Basic badge system implementation
    - Firebase integration for scores
    - Advanced AI bias mitigation (in progress)
    - Dynamic weighting system
3. **Job Recommendation Engine - 60% Complete**
    - Basic semantic matching

- Job listing integration
- Firebase data structure
- Advanced ML ranking (in progress)
- Real-time updates

Technical Infrastructure:

- **Frontend (Flutter):** 75% Complete
  - Authentication screens
  - Profile management
  - Job browsing interface
  - Mock interview UI (in progress)
- **Backend (Firebase):** 80% Complete
  - Authentication system
  - Firestore database structure
  - Cloud Functions setup
  - Cloud Storage configuration

Phase 3: Advanced Modules & Integration (PLANNED - 0%)

Status: Not Started
Scheduled: Weeks 17-24 (Feb-Mar 2026)

Phase 4: Final Integration & Deployment (PLANNED - 0%)

Status: Not Started
Scheduled: Weeks 25-32 (Apr-May 2026)

7. References

[1] Rozee.pk (2025). Pakistan's No.1 Job Portal. https://www.rozee.pk
[2] LinkedIn (2025). Professional Networking Platform. https://www.linkedin.com
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[5] Udemy (2025). Skill Development Platform. https://www.udemy.com
[6] spaCy (2025). Industrial-Strength Natural Language Processing Library. https://spacy.io
[7] NLTK (2025). Natural Language Toolkit for Text Processing. https://www.nltk.org
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[9] scikit-learn (2025). Machine Learning Library for Python. https://scikit-learn.org
[10] PyTorch (2025). Deep Learning Framework. https://pytorch.org
[11] TensorFlow (2025). Open Source ML Platform. https://www.tensorflow.org
[12] Flutter (2025). Cross-Platform App Development Framework. https://flutter.dev
[13] React Native (2025). Cross-Platform Mobile Development Framework. https://reactnative.dev
[14] TestGorilla (2025). Pre-Employment Screening & Assessment Tools. https://www.testgorilla.com

Appendix A: Glossary

- **AI**: Artificial Intelligence, enabling automated decision-making and problem-solving.
- **NLP:** Natural Language Processing, AI techniques for analyzing and understanding human language.
- **Hassle Free:** AI-POWERED CAREER ENHANCEMENT AND RECRUITMENT PLATFORM
- **Resume Parsing:** Extracting structured information (skills, education, experience) from resumes.
- **Firebase:** Cloud platform providing database, authentication, and serverless services.
- **Mock Interview:** AI-driven simulated interview assessing candidate skills and providing feedback.
- **Role-based Access Control:** Security method granting permissions based on user roles.
- **RESTful API:** Web interface using HTTP to access and manipulate system data.
- **TLS/SSL:** Encryption protocols for secure data transmission.

Recommendation Engine**: AI module matching user profiles with suitable job listings**

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 1: List of non-trivial defects

This document has been adapted from the following:

1. Previous project templates at UCP
2. High-level Technical Design, Centers for Medicare & Medicaid Services. (www.cms.gov)